

Braid Monodromy of Analytic Matrix Families: Sharp Discriminant Constants and Torus Extensions

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Abstract

To an analytic family of $n \times n$ matrices with generically simple spectrum, each loop in the parameter space assigns a conjugacy class in the braid group B_n via the eigenvalue monodromy. We develop a finite, operator-norm certificate for this assignment: a positive discriminant margin on a compact based graph, together with a sharp Lipschitz constant for the characteristic discriminant, yields an explicit perturbation radius within which every generator braid and every relation among them is stable.

The sharp Lipschitz constants are computed exactly in the quadratic and traceless cubic cases. The exact operator-norm Lipschitz constant for the discriminant is $8R$ on the 2×2 ball and $162R^5$ on the traceless 3×3 ball of radius R ; both are attained by unit-root diagonal families. A roots-of-unity identity shows that the pattern $n^{n+1}(n-1)R^{n(n-1)-1}$, proved here for $n \in \{2, 3\}$, is the natural candidate for all n . For $n = 4$ we prove the sharp constant $3072R^{11}$: the non-normal rank-one defect case is closed by a finite four-point Blaschke–Pick certificate and a blow-up resolution of its only hard Bernstein corner.

Three families illustrate the certificate with complete braid computations. For the companion cubic $E^3 - 3E - k$, the two-puncture monodromy representation surjects onto B_3 with kernel the normal closure of the Artin relation. For $E^3 - k^3 - 1$, the image is the cyclic group $\langle \sigma_1 \sigma_2 \rangle$. On the two-dimensional Brillouin torus $\mathbb{T}^2$, we prove that families of the form $H = g(k_1, k_2)M$ with $M = \text{diag}(1, \omega, \omega^2)$ always have braid monodromy valued in the centre $Z(B_3)$: rigid rotation of the eigenvalue triangle forces every loop to map to a power of the full twist $(\sigma_1 \sigma_2)^3$. Explicit Zariski–van Kampen computations determine the representations on $\pi_1(\mathbb{T}^2)$ and $\pi_1(\mathbb{T}^2 \setminus C)$ for specific two-parameter families, and a quartic A_3 arrangement provides a B_4 -valued example with explicit certified radius.

Keywords: braid monodromy; exceptional point; discriminant complement; matrix family; configuration space; non-Hermitian matrix; combinatorial spectral topology; Nevanlinna–Pick interpolation; certified computation.

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1 Introduction

A matrix loop with simple spectrum determines a braid of its eigenvalues. A multi-parameter family determines a representation

$$\beta_H : \pi_1(\mathcal{P} \setminus \mathcal{D}, p_0) \longrightarrow B_n$$

from the complement of the repeated-eigenvalue locus. The aim of this paper is to make such representations certifiable when the family is known only through finitely many matrix samples on selected loops.

The question is natural in non-Hermitian spectral theory. Exceptional points are eigenvalue branch points, and loops around them carry braid charges [12, 17, 2, 10, 9, 19, 18]. Thus a physical spectral motion is converted into a word in an Artin braid group, while the arrangement of exceptional points supplies the discriminant complement on which the word is read. Numerically, however, the loop is a mesh, not a continuum. One has to know that the graph stays off the discriminant between samples, and that small matrix errors do not change the braid word or the relations among several chosen loops.

The certificate uses only elementary homotopy and a Lipschitz estimate. The unordered spectrum maps the discriminant complement to $\text{UConf}_n(\mathbb{C})$. On a compact based graph G , a positive margin M_G for the characteristic discriminant keeps the straight-line homotopy from a family to a nearby family inside configuration space, provided the perturbation is smaller than the associated operator-norm radius. Hence the generator braids carried by G are fixed simultaneously. If only a mesh is available, the observed margin is reduced by the explicit correction $\mathcal{G}(R_*)L_G h_G$.

Section 2 fixes the configuration-space conventions. Section 3 proves the quantitative input used later: for trace-zero cubic matrices $\mathcal{G}_{\text{tr}=0,3}(R) = 162R^5$, while for trace-zero quartic matrices it proves the sharp constant $3072R^{11}$. The non-normal quartic upper bound is reduced to, and closed by, the four-point Nevanlinna–Pick certificate. The graph, sampling, and adaptive gap certificates occupy Sections 4–6. The examples separate two cubic arrangements with the same local degree-three flavour but different global monodromy. The companion cubic $E^3 - 3E - k$ gives a surjection onto B_3 , whereas the factorised family $E^3 - k^3 - 1$ has cyclic image generated by $\sigma_1\sigma_2$. Section 9 relates these computations to the cusp and to a quartic A_3 benchmark. Section 10 extends the graph certificate to the two-dimensional Brillouin torus $\mathbb{T}^2$: the discriminant locus is generically a finite set of isolated exceptional points (complex-valued Δ) or an embedded curve (real-valued Δ), and two versions of the stability theorem cover the cases $M_{\mathbb{T}^2} > 0$ and $C \neq \emptyset$ separately. Propositions 10.4–10.5 work out explicit families $H = gM$ whose braid monodromy maps $\pi_1(\mathbb{T}^2)$ and $\pi_1(\mathbb{T}^2 \setminus C)$ into the centre $Z(B_3)$, with complete Zariski–van Kampen verification.

2 Configuration spaces and spectral braid monodromy

Let

$$\text{Conf}_n(\mathbb{C}) = \{(z_1, \dots, z_n) \in \mathbb{C}^n : z_i \neq z_j \text{ for } i \neq j\}$$

be the ordered configuration space and let

$$\text{UConf}_n(\mathbb{C}) = \text{Conf}_n(\mathbb{C})/S_n$$

be the unordered configuration space. The braid group is

$$B_n = \pi_1(\text{UConf}_n(\mathbb{C}), Q_0),$$

where Q_0 is a chosen unordered base configuration. Equivalently, a loop in $\text{UConf}_n(\mathbb{C})$ is an unordered motion of n distinct points in the complex plane, and its homotopy class is a braid.

Let $\mathcal{P}$ be a connected smooth parameter domain and let

$$H : \mathcal{P} \longrightarrow M_n(\mathbb{C})$$

be continuous. Write

$$P_p(E) = \det(EI - H(p))$$

and denote by

$$\Delta_H(p) = \text{Disc}(P_p)$$

the classical polynomial discriminant. The discriminant locus is

$$\mathcal{D}_H = \{p \in \mathcal{P} : \Delta_H(p) = 0\}.$$

We use the standard resultant/discriminant convention; see, for example, [11] for background. On

$$X_H = \mathcal{P} \setminus \mathcal{D}_H$$

the spectrum consists of n distinct points, so the unordered eigenvalue map

$$\Lambda_H : X_H \longrightarrow \text{UConf}_n(\mathbb{C}), \quad \Lambda_H(p) = \text{spec } H(p)$$

is continuous.

Definition 2.1 (Spectral braid monodromy). *Fix $p_0 \in X_H$ and set $Q_0 = \Lambda_H(p_0)$. The spectral braid monodromy representation of H is*

$$\beta_H = (\Lambda_H)_* : \pi_1(X_H, p_0) \longrightarrow B_n.$$

This definition uses unordered eigenvalues and therefore does not require a global labeling. If one lifts locally to ordered eigenvalue branches, then the associated permutation monodromy is the image of β_H under the natural map $B_n \rightarrow S_n$.

Remark 2.2 (Basepoint and conjugacy). *Changing the basepoint inside the same connected component conjugates the representation by the braid obtained along the connecting path. Changing an initial ordering of the eigenvalues also conjugates the ordered description. Thus the intrinsic object is the simultaneous conjugacy class of the braid monodromy representation.*

Convention 2.3 (Upper-half-plane meridians). *When a local branch point is approached from a real basepoint and two ordered real eigenvalues collide, a positive parameter meridian is drawn so that the corresponding two eigenvalue branches pass through the upper half of the eigenvalue plane. With this convention the positive half-twist of the adjacent strands in positions $i, i+1$ is denoted by the Artin generator σ_i ; reversing the meridian orientation gives σ_i^{-1} . This is the convention used in the explicit root-tracking tables below.*

Lemma 2.4 (Homotopy invariance). *Let $H_s : X \rightarrow M_n(\mathbb{C})$, $s \in [0, 1]$, be a continuous family on a connected space X such that $\text{Disc}(H_s(x)) \neq 0$ for every $(s, x) \in [0, 1] \times X$. Then the maps $\Lambda_{H_0}, \Lambda_{H_1} : X \rightarrow \text{UConf}_n(\mathbb{C})$ are homotopic. Consequently they induce the same braid monodromy representation up to the standard basepoint conjugacy.*

Proof. The map $(s, x) \mapsto \Lambda_{H_s}(x)$ is a homotopy through $\text{UConf}_n(\mathbb{C})$ because the discriminant never vanishes. Homotopic maps induce the same homomorphism on fundamental groups after identifying basepoints by the induced path in the target. $\square$

Remark 2.5. *Analyticity of H is not needed for the topological construction on the complex plane. Analyticity is useful for describing the local structure of $\mathcal{D}_H$ and for exceptional-point expansions, as in the standard local perturbation theory of eigenvalues [15]; the stability certificate below only uses continuity, a norm bound, and a Lipschitz estimate for the discriminant.*

3 Sharp discriminant Lipschitz constants

The graph certificates below require an operator-norm Lipschitz constant for the discriminant on the relevant matrix class. This section computes these constants exactly for $n = 2$ and $n = 3$ (traceless), gives the current traceless $n = 4$ bounds and reduction, and states the general conjecture. The cubic examples in Sections 7 and 8 use the exact constant $162R^5$; the quartic example in Section 9 uses the $n = 4$ bounds.

Let

$$\mathcal{T}_3(R) = \{H \in M_3(\mathbb{C}) : \operatorname{tr} H = 0, \|H\|_2 \leq R\}.$$

For $H \in \mathcal{T}_3(R)$, write

$$\det(EI - H) = E^3 + pE + q, \quad p = -\frac{1}{2} \operatorname{tr}(H^2), \quad q = -\frac{1}{3} \operatorname{tr}(H^3),$$

so that

$$\operatorname{Disc}(H) = -4p^3 - 27q^2.$$

Theorem 3.1 (Exact trace-zero cubic discriminant constant). *For $R > 0$, the exact operator-norm Lipschitz constant of the discriminant on $\mathcal{T}_3(R)$ is*

$$\mathcal{G}_{\operatorname{tr}=0,3}(R) = 162R^5.$$

Equivalently, for all $A, B \in \mathcal{T}_3(R)$,

$$|\operatorname{Disc}(A) - \operatorname{Disc}(B)| \leq 162R^5 \|A - B\|_2,$$

and the coefficient $162R^5$ is sharp.

Proof. See Appendix G. □

Remark 3.2 (Why the structured constant matters). *The unstructured cubic ball admits coefficient-box constants that are much larger. Theorem 3.1 is usable here because trace-zero is a convex linear constraint and is preserved by the straight-line homotopies used in Theorem 4.2. Thus, for trace-preserving perturbations of the cubic examples below, all graph-level radii may use $162R_*^5$ rather than a general cubic box constant.*

Proposition 3.3 (Two-dimensional Lipschitz constant). *For $n = 2$ and $R > 0$, the exact operator-norm Lipschitz constant of the discriminant on $\{\|H\|_2 \leq R\}$ is $\mathcal{G}_2(R) = 8R$, attained by $J_2 = \operatorname{diag}(1, -1)$. The coefficient-route value $16R$ exceeds it by a factor of 2.*

Proof. Write $B = H - \frac{1}{2}(\operatorname{tr} H)I$; then $\operatorname{Disc}(H) = 2\operatorname{tr}(B^2)$ and $\nabla_H \operatorname{Disc} = 4B^T$. Von Neumann gives $|D \operatorname{Disc}(H)[\delta H]| \leq 4\|B\|_* \|\delta H\|_2 \leq 8\|H\|_2 \|\delta H\|_2 \leq 8R \|\delta H\|_2$. The lower bound $\mathcal{G}_2(R) \geq 8R$ follows from the radial family tJ_2 via homogeneity. □

Lemma 3.4 (Roots-of-unity identity). *Let $n \geq 2$, $\omega = e^{2\pi i/n}$, $J_n = \operatorname{diag}(1, \omega, \dots, \omega^{n-1})$. Then $\operatorname{tr} J_n = 0$, $\|J_n\|_2 = 1$, $|\operatorname{Disc}(J_n)| = n^n$, and*

$$\sum_{j=1}^{n-1} \frac{1}{1 - \omega^j} = \frac{n-1}{2}.$$

Consequently the lower bound $\mathcal{G}_n^{\operatorname{tr}=0}(R) \geq n(n-1) |\operatorname{Disc}(J_n)| R^{n(n-1)-1}$ gives

$$\mathcal{G}_n^{\operatorname{tr}=0}(R) \geq n^{n+1} (n-1) R^{n(n-1)-1}. \quad (3.1)$$

This is exact for $n = 2$ ($8R$) and $n = 3$ ($162R^5$).

Proof. Trace vanishes since $\sum_{j=0}^{n-1} \omega^j = 0$. All eigenvalues have modulus 1, so $\|J_n\|_2 = 1$. The discriminant of $z^n - 1$ equals n^n in absolute value. For the identity, pair j with $n-j$: $\frac{1}{1-\omega^j} + \frac{1}{1-\omega^{-j}} = 1$, and summing over $j = 1, \dots, n-1$ gives $(n-1)/2$. Substituting into the radial lower bound yields (3.1). $\square$

Theorem 3.5 (Traceless 4×4 bounds and reduced sharpness criterion). *Let $\mathcal{T}_4(R) = \{H \in M_4(\mathbb{C}) : \text{tr } H = 0, \|H\|_2 \leq R\}$.*

$$3072R^{11} \leq \mathcal{G}_4^{\text{tr}=0}(R) \leq \frac{60416}{3}R^{11}.$$

The lower bound is attained in the radial limit by tJ_4 , $J_4 = \text{diag}(1, i, -1, -i)$. If the reduced four-point Pick matrix

$$\left[\frac{1 - r_i \overline{r_j}}{1 - \lambda_i \overline{\lambda_j}} \right]_{i,j=1}^4, \quad r_i = 768^{-1}(c_1 + c_2 \lambda_i + c_3 \lambda_i^2),$$

is positive semidefinite for every centred $\lambda_j \in \overline{\mathbb{D}}$, then the sharp value is

$$\mathcal{G}_4^{\text{tr}=0}(R) = 3072R^{11}.$$

Proof. Lower bound. J_4 satisfies $\text{tr } J_4 = 0$, $\|J_4\|_2 = 1$, $P_{J_4}(E) = E^4 - 1$, $|\text{Disc}(J_4)| = 256$. By Lemma 3.4, $\mathcal{G}_4^{\text{tr}=0}(R) \geq 4 \cdot 3 \cdot 256 \cdot R^{11} = 3072R^{11}$.

A priori upper bound. Newton's identities give $|a_2| \leq 2R^2$, $|a_1| \leq \frac{4}{3}R^3$, and $|a_0| \leq R^4$ for $P_H(E) = E^4 + a_2E^2 + a_1E + a_0$. Bounding each partial derivative of the quartic discriminant and using Cayley–Hamilton, $\text{adj } H = -H^3 - a_2H - a_1I$, with $\|H^k\|_* \leq 4R^k$, gives $\|\nabla_H \text{Disc}\|_* \leq \frac{60416}{3}R^{11}$.

Reduced sharp upper bound. By homogeneity it suffices to take $R = 1$. The normal strata are handled by (3.2) and the subsequent unit-circle and mixed-stratum reductions. For non-normal matrices, Schur triangularisation and the finite Blaschke model reduce the problem to the rank-one defect model S_B . By Proposition 3.20, the reduced Pick matrix is positive semidefinite on the centered four-point disk. Identity (3.9) therefore implies positivity of the original Pick matrix (3.8). The finite Nevanlinna–Pick theorem and Sarason's lifting theorem then give $\|c_1S_B + c_2S_B^2 + c_3S_B^3\|_2 \leq 768$, hence

$$\|c_1S_B + c_2S_B^2 + c_3S_B^3\|_* \leq 4\|c_1S_B + c_2S_B^2 + c_3S_B^3\|_2 \leq 3072.$$

Scaling gives the factor R^{11} . $\square$

Conjecture 3.6 (Exact traceless Lipschitz constant). *For every $n \geq 2$ and $R > 0$,*

$$\mathcal{G}_n^{\text{tr}=0}(R) = n^{n+1}(n-1)R^{n(n-1)-1},$$

attained by tJ_n . The cases $n = 2, 3, 4$ are proved here; the formula remains conjectural for $n \geq 5$.

Remark 3.7 (Proof mechanism for Conjecture 3.6 at $n = 4$). *For the traceless quartic operator-norm ball, the normal part is handled by root stratification and the non-normal part is reduced to a finite Blaschke-model Pick certificate.*

Gradient formula. *For $H \in \mathcal{T}_4(R)$ with characteristic polynomial $P_H(E) = E^4 + a_2E^2 + a_1E + a_0$, the quartic discriminant is*

$$\Delta = 256a_0^3 - 128a_2^2a_0^2 + 144a_2a_1^2a_0 - 27a_1^4 + 16a_2^4a_0 - 4a_2^3a_1^2.$$

Using the Cayley–Hamilton identity $\text{adj } H = -(H^3 + a_2H + a_1I)$ and the traceless constraint (which kills the I -term against traceless test matrices), one computes

$$D\Delta(H)[X] = \text{tr}(A(H)X), \quad A(H) = c_1H + c_2H^2 + c_3H^3,$$

where

$$c_1 = -(\Delta_{a_2} + a_2\Delta_{a_0}), \quad c_2 = -\Delta_{a_1}, \quad c_3 = -\Delta_{a_0},$$

and $\Delta_{a_j} = \partial\Delta/\partial a_j$. Hence $\mathcal{G}_4^{\text{tr}=0}(R) = \sup\{\|A(H)\|_* : H \in \mathcal{T}_4(R)\}$.

For normal diagonal $H = \text{diag}(\lambda_j)$ the nuclear norm is $\|A(H)\|_* = \sum_j |\phi_H(\lambda_j)|$, where $\phi_H(z) = c_1z + c_2z^2 + c_3z^3$. Using the discriminant product formula $\partial_{\lambda_j}\Delta = \Delta \cdot 2 \sum_{k \neq j} (\lambda_j - \lambda_k)^{-1}$, the normal target reduces to

$$F(\lambda) := |\Delta(\lambda)| \left| \sum_{j=1}^4 \frac{1}{\lambda_j - \lambda_k} \right| \leq 3072, \quad |\lambda_j| \leq 1, \quad \sum_j \lambda_j = 0. \quad (3.2)$$

At $\lambda = \{1, i, -1, -i\}$: $|\Delta| = 256$ and $\sum_j |L_j| = 12$, giving $F = 3072$.

Stratum 1: all four roots on $\partial\mathbb{D}$. The constraint $|\lambda_j| = 1$, $\sum_j \lambda_j = 0$ forces roots to come in opposite pairs $\{\alpha, -\alpha, \beta, -\beta\}$. By rotation invariance set $\alpha = 1$, $\beta = u$, $|u| = 1$, so $\lambda = \{1, -1, u, -u\}$. Writing $u^2 = e^{i\theta}$, a direct computation gives

$$F = 32|1 - u^2|^3(|5 - u^2| + |5u^2 - 1|).$$

This is a one-variable inequality in θ ; the maximum over $\theta \in [0, 2\pi)$ is 3072, attained at $\theta = \pi/2$ (i.e. $u = \pm i$), which is the square spectrum. Hence $F \leq 3072$ on Stratum 1.

Stratum 2: pair-symmetric mixed family. Take $\lambda = \{1, -1, u, -u\}$ with $|u| = r \leq 1$. Set $x = r^2$, $y = \cos(2 \arg u)$. With

$$A = 1 + x^2 - 2xy, \quad B = 25 + x^2 - 10xy, \quad C = 25x^2 + 1 - 10xy,$$

the inequality $F \leq 3072$ is equivalent to

$$xA^{3/2}(B^{1/2} + x^{-1/2}C^{1/2}) \leq 96. \quad (3.3)$$

Since A, B, C are each decreasing in y , the left side is maximised at $y = -1$, giving $A = (1 + x)^2$, $B = (5 + x)^2$, $C = (5x + 1)^2$. Substituting $t = \sqrt{x} \in [0, 1]$, the left side of (3.3) becomes $t^2(1 + t^2)^3 \cdot \frac{1}{t}(t^3 + 5t^2 + 5t + 1)$, which equals $t(1 + t^2)^3(1 + t)^3$. This is increasing on $[0, 1]$ and equals 96 at $t = 1$. Hence $F \leq 3072$ on Stratum 2, with equality only at $r = 1$, $y = -1$, i.e. the square spectrum.

Stratum 3: the (3+1) normal stratum. The only normal stratum not covered by the two preceding algebraic estimates is the mixed case with three roots on the unit circle and one root in the disk. After rotation it has the following explicit form:

$$\lambda_4 = -r, \quad 0 \leq r < 1, \quad \lambda_1 + \lambda_2 + \lambda_3 = r, \quad \lambda_1\lambda_2\lambda_3 = e^{i\chi}.$$

The three boundary roots are the roots of

$$Q_{r,\chi}(z) = z^3 - rz^2 + re^{i\chi}z - e^{i\chi},$$

and

$$P(z) = (z + r)Q_{r,\chi}(z).$$

A direct resultant computation gives

$$|\Delta(P)| = |r^4 + 18r^2 - 27 - 8r^3 \cos \chi| |2r^3 + e^{i\chi}(1 + r^2)|^2. \quad (3.4)$$

Moreover the logarithmic force at the interior root is

$$L_{-r} = 2 \sum_{Q_{r,\chi}(\lambda)=0} \frac{1}{-r - \lambda} = -2 \frac{5r^2 + re^{i\chi}}{2r^3 + e^{i\chi}(1 + r^2)}. \quad (3.5)$$

Equations (3.4)–(3.5), together with the corresponding formula

$$L_\lambda = 2 \left(\frac{Q''_{r,\chi}(\lambda)}{2Q'_{r,\chi}(\lambda)} + \frac{1}{\lambda + r} \right), \quad Q_{r,\chi}(\lambda) = 0,$$

reduce the (3+1) problem to the explicit two-variable inequality

$$F(r, \chi) = |\Delta(P)| \left(|L_{-r}| + \sum_{Q_{r,\chi}(\lambda)=0} |L_\lambda| \right) \leq 3072.$$

This formulation also corrects a tempting but false monotonicity route: for fixed χ , $F(r, \chi)$ is not generally increasing in r , and $F(0, \chi) = 324$, not 0.

We now close this stratum by replacing the exact sum over the three unit roots by a sharp enough algebraic upper bound. Put $c = \cos \chi$ and write

$$D(r, c) = (27 - 18r^2 - r^4 + 8r^3c)(4r^6 + (1 + r^2)^2 + 4r^3(1 + r^2)c),$$

$$I(r, c) = \frac{4r^2(25r^2 + 1 + 10rc)}{4r^6 + (1 + r^2)^2 + 4r^3(1 + r^2)c}.$$

Thus $D = |\Delta(P)|$ and $I = |L_{-r}|^2$ on $0 \leq r \leq 1$, $-1 \leq c \leq 1$. A resultant calculation using $\bar{\lambda} = 1/\lambda$ for the three unit roots shows that the three numbers $U_\lambda = |L_\lambda|^2$ are the roots of a cubic

$$a(r, c)U^3 + b(r, c)U^2 + \dots = 0,$$

where

$$a(r, c) = (-8cr^3 + r^4 + 18r^2 - 27)^2(4cr^5 + 4cr^3 + 4r^6 + r^4 + 2r^2 + 1),$$

$$b(r, c) = -4(-8cr^3 + r^4 + 18r^2 - 27)B(r, c),$$

and

$$B(r, c) = -80c^2r^8 - 160c^2r^6 - 40cr^9 + 142cr^7 - 180cr^5 - 786cr^3$$

$$+ 4r^{10} + 155r^8 - 91r^6 + r^4 - 369r^2 - 324.$$

Consequently

$$\sum_{Q_{r,\chi}(\lambda)=0} |L_\lambda| \leq \sqrt{3 \sum_{\lambda} |L_\lambda|^2} = \sqrt{3S_1(r, c)}, \quad S_1 = -\frac{b}{a}.$$

It is therefore enough to prove

$$\mathcal{B}(r, c) := D(r, c)(\sqrt{3S_1(r, c)} + \sqrt{I(r, c)}) \leq 3072. \quad (3.6)$$

First, $\mathcal{B}$ is increasing in c . If $A = 3S_1$ and $Y = I$, then

$$2\partial_c \mathcal{B} = \left(2D_c + D \frac{A_c}{A} \right) \sqrt{A} + \left(2D_c + D \frac{Y_c}{Y} \right) \sqrt{Y}.$$

The second coefficient has numerator

$$R(r, c) = 640c^2r^8 + 640c^2r^6 + 1640cr^9 + 608cr^7 + 648cr^5 + 1200cr^3 \\ + 730r^{10} - 1085r^8 + 1260r^6 + 1534r^4 + 242r^2 + 135,$$

and denominator $10cr + 25r^2 + 1 \geq 0$. The first coefficient is $2r^3N(r, c)/E(r, c)$, where

$$E(r, c) = -80c^2r^8 - 160c^2r^6 - 40cr^9 + 142cr^7 - 180cr^5 - 786cr^3 \\ + 4r^{10} + 155r^8 - 91r^6 + r^4 - 369r^2 - 324,$$

and

$$N(r, c) = -5120c^3r^{13} - 15360c^3r^{11} - 10240c^3r^9 - 5280c^2r^{14} + 6336c^2r^{12} \\ + 8256c^2r^{10} - 72768c^2r^8 - 65568c^2r^6 - 672cr^{15} + 18264cr^{13} \\ - 10088cr^{11} - 23184cr^9 + 47504cr^7 - 91848cr^5 - 105864cr^3 \\ + 136r^{16} + 3210r^{14} - 13023r^{12} + 21960r^{10} + 20763r^8 \\ - 28910r^6 + 4087r^4 - 4640r^2 - 29403.$$

For $w = (1 + c)/2 \in [0, 1]$, the Bernstein coefficients in w of R , $-E$, and $-N$ are one-variable polynomials in r . They are listed in Appendix D. After the displayed endpoint factors are removed, Sturm root counts show that every remaining factor has no zero in $(0, 1)$; equivalently every Bernstein coefficient is non-negative on $0 \leq r \leq 1$. The endpoint values are

	R	$-E$	$-N$
$w = 0$	$135 \rightarrow 0$	$324 \rightarrow 0$	$29403 \rightarrow 0$
middle coefficients	$135 \rightarrow 1536$	$324 \rightarrow 384$	$29403 \rightarrow 0, 29403 \rightarrow 49152$
$w = 1$	$135 \rightarrow 8192$	$324 \rightarrow 1728$	$29403 \rightarrow 393216$

and the Sturm count in $(0, 1)$ is zero for each listed coefficient (endpoint zeros occur only at $r = 1$ in the first column entries indicated by “ $\rightarrow 0$ ”). Thus $R \geq 0$, $E \leq 0$, $N \leq 0$, and hence $\partial_c \mathcal{B} \geq 0$.

It remains to prove $\mathcal{B}(r, 1) \leq 3072$. At $c = 1$,

$$D = (3 - r)^3(1 + r)^3(2r^2 - r + 1)^2, \\ I = \frac{4r^2(5r + 1)^2}{(1 + r)^2(2r^2 - r + 1)^2}, \\ S_1 = \frac{4(2r^5 - 9r^4 - 24r^3 - 17r^2 - 24r - 36)}{(r - 3)(1 + r)^2(2r^2 - r + 1)}.$$

Let $T = 3072/D$. The implication $\sqrt{3S_1} + \sqrt{I} \leq T$ follows from

$$Q := T^2 - 3S_1 - I \geq 0, \quad P := Q^2 - 12S_1I \geq 0.$$

Direct simplification gives

$$Q = \frac{-8q_{20}(r)}{(r - 3)^6(1 + r)^6(2r^2 - r + 1)^4},$$

where $q_{20}(0) = -1166526$, $q_{20}(1) = -442368$, and q_{20} has no root in $(0, 1)$ by Sturm's theorem. Hence $Q > 0$. Similarly

$$P = \frac{256(r - 1)p_{19}(r)p_{20}(r)}{(r - 3)^{12}(1 + r)^{12}(2r^2 - r + 1)^8},$$

where $p_{19}(0) = 583263$, $p_{19}(1) = 2433024$, $p_{20}(0) = -583263$, $p_{20}(1) = -589824$, and both p_{19}, p_{20} have no root in $(0, 1)$. Thus $p_{19} > 0$, $p_{20} < 0$, and $P \geq 0$ for $0 \leq r \leq 1$. Therefore $\mathcal{B}(r, 1) \leq 3072$, with equality only at $r = 1$. Since $F \leq \mathcal{B}$ and $\mathcal{B}$ is increasing in c ,

$$F(r, \chi) \leq 3072$$

on the whole $(3+1)$ normal stratum, with equality only in the boundary limit giving the square spectrum.

Non-normal case. We now pass from normal contractions to arbitrary traceless contractions. The argument uses the complex extreme boundary of the trace-zero contraction ball. For fixed trace-zero direction, the map $H \mapsto A(H) = c_1 H + c_2 H^2 + c_3 H^3$ is holomorphic and $\|A(H)\|_*$ is plurisubharmonic, since

$$\|A(H)\|_* = \sup_{\|Y\| \leq 1} |\operatorname{tr}(Y^* A(H))|.$$

Hence its maximum on the compact trace-zero contraction ball is attained on the complex extreme boundary.

Let T be such a complex extreme contraction in Schur form. If $D_L = (I - TT^*)^{1/2}$ and $D_R = (I - T^*T)^{1/2}$, then every sufficiently small perturbation of the form $D_L Z D_R$ keeps T inside the contraction ball to first order, and the standard Julia-operator parametrisation makes this an actual analytic disk. The trace-zero slice removes only the single complex linear condition

$$\operatorname{tr}(D_L Z D_R) = 0.$$

Therefore, if $\operatorname{rank} D_L \operatorname{rank} D_R \geq 2$, there is a non-zero trace-zero analytic disk through T , contradicting complex extremality. Since T is square, $\operatorname{rank} D_L = \operatorname{rank} D_R$. Thus a non-unitary complex extreme point has defect rank one:

$$\operatorname{rank}(I - T^*T) = \operatorname{rank}(I - TT^*) = 1.$$

The unitary extreme points are normal and have already been covered by the all-boundary normal estimate.

A finite-dimensional completely non-unitary contraction with defect indices $(1, 1)$ is unitarily equivalent to the compressed shift S_B associated with a finite Blaschke product

$$B(z) = \prod_{j=1}^4 \frac{z - \lambda_j}{1 - \bar{\lambda}_j z}.$$

In the Takenaka–Malmquist basis,

$$(S_B)_{ii} = \lambda_i, \quad (S_B)_{ij} = \left(\prod_{k=i+1}^{j-1} (-\lambda_k) \right) (1 - |\lambda_i|^2)^{1/2} (1 - |\lambda_j|^2)^{1/2}, \quad i < j.$$

Thus all genuinely non-normal entries carry defect factors. Put $A_B = c_1 S_B + c_2 S_B^2 + c_3 S_B^3 = \phi_\lambda(S_B)$. The entries are explicit. The diagonal entries are $(A_B)_{ii} = \phi(\lambda_i)$, the first upper diagonal is

$$(A_B)_{i,i+1} = d_i d_{i+1} [c_1 + c_2(\lambda_i + \lambda_{i+1}) + c_3(\lambda_i^2 + \lambda_i \lambda_{i+1} + \lambda_{i+1}^2)],$$

where $d_i = (1 - |\lambda_i|^2)^{1/2}$, and the two longer upper diagonals are obtained by expanding $c_1 S_B + c_2 S_B^2 + c_3 S_B^3$. Each such entry is $d_i d_j$ times an explicit polynomial in the λ 's and the defects.

The structural reduction therefore shows that Conjecture 3.6 for $n = 4$ follows from the finite contraction inequality

$$\|A_B\|_2 \leq 768, \quad |\lambda_j| \leq 1, \quad \sum_j \lambda_j = 0, \quad (3.7)$$

because $\|A_B\|_* \leq 4\|A_B\|_2 \leq 3072$. Equivalently, the Frobenius certificate $\|A_B\|_F \leq 1536$ follows from (3.7). The cells with all defects zero reduce to the normal estimate already proved; the cells with exactly one non-zero defect reduce to the (3+1) certificate above. The remaining interior defect cells constitute a compact semialgebraic set parameterised by $\lambda_j = x_j + iy_j$ and $u_j = 1 - |\lambda_j|^2 \geq 0$, and (3.7) is equivalent there to the finite Nevanlinna–Pick positivity certificate below.

Remark 3.8 (Finite Pick positivity certificate for the Blaschke model). *Let B be the finite Blaschke product with zeros $\lambda_1, \dots, \lambda_4 \in \mathbb{D}$, counted with multiplicity, and let S_B be the compressed shift in the Takenaka–Malmquist basis above. Set*

$$q_\lambda(z) = 768^{-1} \phi_\lambda(z).$$

If $\sum_j \lambda_j = 0$, then the Pick matrix

$$\Pi_\lambda = \left[\frac{1 - q_\lambda(\lambda_i) \overline{q_\lambda(\lambda_j)}}{1 - \lambda_i \overline{\lambda_j}} \right]_{i,j=1}^4 \quad (3.8)$$

is the finite positivity object whose semidefiniteness gives the non-normal quartic sharp upper bound; this semidefiniteness is proved in Proposition 3.20.

Lemma 3.9 (Reduced Pick recursion). *By the finite Nevanlinna–Pick theorem, $\Pi_\lambda \succeq 0$ is equivalent to the existence of a Schur function $f \in H^\infty(\mathbb{D})$, $\|f\|_\infty \leq 1$, with*

$$f(\lambda_j) = q_\lambda(\lambda_j), \quad j = 1, \dots, 4.$$

Sarason’s commutant lifting theorem would then give

$$\|q_\lambda(S_B)\|_2 = \inf\{\|f\|_\infty : f(\lambda_j) = q_\lambda(\lambda_j)\} \leq 1.$$

Thus Remark 3.8 is exactly the remaining finite certificate needed for the non-normal sharp $n = 4$ upper bound.

Moreover the Pick problem has one further exact reduction. Since $q_\lambda(0) = 0$, write

$$q_\lambda(\lambda_i) = \lambda_i r_i, \quad r_i = 768^{-1} (c_1 + c_2 \lambda_i + c_3 \lambda_i^2).$$

Then

$$\frac{1 - \lambda_i r_i \overline{\lambda_j r_j}}{1 - \lambda_i \overline{\lambda_j}} = 1 + \lambda_i \overline{\lambda_j} \frac{1 - r_i \overline{r_j}}{1 - \lambda_i \overline{\lambda_j}}. \quad (3.9)$$

Consequently the finite Pick certificate of Remark 3.8 follows from the reduced finite Pick positivity

$$\left[\frac{1 - r_i \overline{r_j}}{1 - \lambda_i \overline{\lambda_j}} \right]_{i,j=1}^4 \succeq 0. \quad (3.10)$$

Proposition 3.10 (Reduced boundary certificates). *The reduced Pick matrix (3.10) is positive semidefinite on the following sharp boundary strata:*

1. the full paired locus $\{a, -a, b, -b\}$, $|a|, |b| \leq 1$;

2. the symmetric centred 3+1 unit-boundary edge

$$\{1, e^{i\theta}, e^{-i\theta}, -r\}, \quad \cos \theta = (r-1)/2, \quad 0 \leq r \leq 1;$$

3. the radial two-pair edge $\{1, -1, ir, -ir\}$, $0 \leq r \leq 1$.

On the full centred 3+1 unit-boundary stratum

$$\{\xi_1, \xi_2, \xi_3, -(\xi_1 + \xi_2 + \xi_3)\}, \quad |\xi_1| = |\xi_2| = |\xi_3| = 1, \quad |\xi_1 + \xi_2 + \xi_3| \leq 1;$$

the first reduced Schur pivot is non-negative, equivalently $|r_i| \leq 1$ for each node. In the full-matrix boundary cases above, equality can occur only at the square $\{1, i, -1, -i\}$, up to rotation and permutation.

Proof. The paired-locus computation gives reduced Schur parameters

$$A, \quad 0, \quad -(B-A)/(1-\overline{A}B), \quad 0,$$

with

$$A = \frac{z^2(1-z^2)^3(5-z^2)}{48}, \quad B = \frac{(1-z^2)^3(1-5z^2)}{48},$$

after rotation to $\{1, -1, z, -z\}$. The estimates $|A| \leq 1$, $|B| \leq 1$ follow from $|1-z^2| \leq 2$ and $|5-z^2|, |1-5z^2| \leq 6$. The symmetric 3+1 and radial two-pair full-matrix cases are the Sturm and factorisation checks recorded in Appendix C. The scalar first-pivot statement on the full 3+1 stratum is the reduced boundary-node certificate in the same appendix: the interior-node inequality is a two-variable Bernstein certificate in $(r, \cos \chi)$, and the unit-node inequality is the dyadic Bernstein certificate in (ρ, y) . $\square$

Proposition 3.11 (First reduced pivot). *For every centred quadruple $\lambda_j \in \overline{\mathbb{D}}$ and every*

$$r_i = 768^{-1}(c_1 + c_2\lambda_i + c_3\lambda_i^2)$$

appearing in (3.10), one has

$$|r_i| \leq 1.$$

Moreover equality can occur only on the three-active boundary strata of Proposition 3.10; in particular the sharp point is the square spectrum, up to rotation and permutation.

Proof. It suffices to treat $i = 1$. Eliminate $\lambda_4 = -(\lambda_1 + \lambda_2 + \lambda_3)$. The compact spectral domain is the analytic polyhedron

$$\mathcal{P} = \{z \in \mathbb{C}^3 : |\ell_j(z)| \leq 1, j = 1, 2, 3, 4\}, \quad \ell_1 = z_1, \ell_2 = z_2, \ell_3 = z_3, \ell_4 = -(z_1 + z_2 + z_3).$$

The quantity r_1 is a holomorphic polynomial in z . Choose a point where $|r_1|$ is maximal. If the active linear forms $\{\ell_j : |\ell_j(z)| = 1\}$ have rank < 3 , there exists $0 \neq v \in \mathbb{C}^3$ annihilated by all active forms. For sufficiently small ζ , the disk $z + \zeta v$ remains in $\mathcal{P}$; hence the maximum-modulus principle forces r_1 to be constant on that disk. Move along the disk until a new linear constraint becomes active. Repeating this finite procedure gives a maximum point at which the active forms have rank 3.

For the four forms above, rank 3 means that at least three roots have modulus one. The centred configurations with three unit roots are exactly the (3+1) boundary edge, with the all-circle case as endpoint, after rotation and permutation. Proposition 3.10 gives positive reduced Pick matrix on that edge; its first Schur pivot is precisely $1 - |r_1|^2$. Therefore $|r_1| \leq 1$. The equality statement follows from the equality statement in Proposition 3.10. $\square$

Lemma 3.12 (Polyhedral maximum principle for centred roots). *Let*

$$\Omega = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : |z_1|, |z_2|, |z_3|, |z_1 + z_2 + z_3| \leq 1\},$$

and set $\lambda_4 = -(z_1 + z_2 + z_3)$. If F is holomorphic in a neighbourhood of Ω , then $\max_{\Omega} |F|$ is attained at a point where at least three of the four constraints $|\lambda_j| \leq 1$ are active.

Proof. Let $z^0 \in \Omega$ be a maximum point and suppose at most two constraints are active. The constraints are given by the four linear functionals

$$\ell_1 = z_1, \quad \ell_2 = z_2, \quad \ell_3 = z_3, \quad \ell_4 = -(z_1 + z_2 + z_3).$$

The common kernel of the active ℓ_j 's has positive complex dimension, so choose $v \neq 0$ in this kernel. Then every active constraint is unchanged on the analytic disk $z^0 + \tau v$, while the inactive constraints remain strictly inside the disk for $|\tau|$ small. Hence $F(z^0 + \tau v)$ has maximal modulus at an interior point of a one-variable disk and is constant there. Moving along this disk until a new constraint becomes active preserves the maximum value. Iterating gives a maximum point with at least three active constraints. $\square$

Lemma 3.13 (Radial boundary blow-up). *Let $\xi_1, \dots, \xi_m \in \partial\mathbb{D}$ be distinct, let $\zeta_1, \dots, \zeta_\ell \in \mathbb{D}$, and prescribe data w_i at the boundary nodes and v_a at the interior nodes. Assume*

$$|w_i| < 1 \quad (1 \leq i \leq m)$$

and assume that the interior Pick matrix

$$\left[\frac{1 - v_a \overline{v_b}}{1 - \zeta_a \overline{\zeta_b}} \right]_{a,b=1}^\ell$$

is positive semidefinite. Then the Pick matrices for the radially perturbed nodes $t\xi_i$, $0 < t < 1$, together with the fixed interior nodes, are positive semidefinite for all t sufficiently close to 1.

Proof. Order the boundary nodes first. The boundary-boundary block has diagonal entries

$$\frac{1 - |w_i|^2}{1 - t^2} \rightarrow +\infty$$

and bounded off-diagonal entries because the ξ_i 's are distinct. Hence this block is positive definite and its inverse tends to zero as $t \uparrow 1$. The Schur complement with respect to this block is the interior Pick matrix plus an $o(1)$ perturbation. If the interior matrix is positive definite, positivity follows for t close to 1. The semidefinite case follows by replacing the interior block by its εI regularisation and letting $\varepsilon \downarrow 0$. $\square$

Proposition 3.14 (Full reduced Pick certificate on the two-angle 3+1 stratum). *The reduced Pick matrix (3.10) is positive semidefinite on the full centred 3+1 stratum*

$$\{\xi_1, \xi_2, \xi_3, -r\}, \quad |\xi_j| = 1, \quad \xi_1 + \xi_2 + \xi_3 = r, \quad 0 \leq r \leq 1.$$

Proof. Away from the square endpoint, the first-pivot certificates in Appendix C give

$$|r(\xi_j)| < 1, \quad j = 1, 2, 3,$$

and

$$|r(-r)| \leq 1.$$

For $r < 1$, the interior block consists only of the node $-r$, and its Pick matrix is the scalar

$$\frac{1 - |r(-r)|^2}{1 - r^2} \geq 0.$$

For the actual radial interior approximants

$$\{t\xi_1, t\xi_2, t\xi_3, -r\}, \quad t < 1,$$

the reduced data vary with t , but continuously:

$$r_t(t\xi_j) \rightarrow r(\xi_j), \quad r_t(-r) \rightarrow r(-r).$$

Thus, for t sufficiently close to 1,

$$1 - |r_t(t\xi_j)|^2 \geq \delta > 0, \quad j = 1, 2, 3.$$

The three boundary diagonal entries of the actual reduced Pick matrix then satisfy

$$\frac{1 - |r_t(t\xi_j)|^2}{1 - t^2} \rightarrow +\infty,$$

while all off-diagonal entries involving distinct unit nodes remain bounded. The Schur complement with respect to this 3×3 block tends to the interior scalar $(1 - |r(-r)|^2)/(1 - r^2) \geq 0$. Hence the approximating Pick matrices are positive semidefinite for t close to 1, and the boundary certificate follows by continuity. The only point where a boundary first pivot can vanish is the square spectrum; that endpoint is already covered by the all-circle/paired certificate in Proposition 3.10. $\square$

Remark 3.15 (Reduced Pick criterion and what is certified). *The anti-holomorphic higher Schur pivots do not need to be estimated directly on the two-angle 3+1 boundary. On that boundary the three unit nodes with strict first-pivot gaps produce positive diagonal blow-up under radial interior approximation, and Lemma 3.13 dominates all higher Schur complements. The genuinely interior rank-one Blaschke cells are closed by Proposition 3.20.*

Proposition 3.16 (Confluent rank-one defect layers). *The reduced Pick certificate is positive semidefinite on the double-root and higher confluent centred strata.*

Proof. Consider first a double-root stratum

$$\{x, x, u, v\}, \quad v = -2x - u, \quad |x|, |u|, |v| \leq 1.$$

For this fixed spectrum the reduced value $r(L)$ is a quadratic polynomial in L , and direct substitution gives

$$r(L) = \frac{(x - u)^3(u + x)^2(u + 3x)^3}{48} (L^2 + xL - (u^2 + 2ux + 2x^2)).$$

Hence

$$\|r\|_{H^\infty(\mathbb{D})} \leq \frac{|(x - u)^3(u + x)^2(u + 3x)^3|}{48} (1 + |x| + |u^2 + 2ux + 2x^2|).$$

The right-hand side is plurisubharmonic in (x, u) on the analytic polyhedron

$$|x| \leq 1, \quad |u| \leq 1, \quad |2x + u| \leq 1,$$

so its maximum is attained on the Shilov boundary. If $|x| = |u| = 1$, then $|2x + u| \leq 1$ forces $u = -x$, and the prefactor vanishes; the same holds for $|x| = |v| = 1$. The only

non-trivial boundary case is therefore $|u| = |v| = 1$. After rotation write $u = e^{i\theta}$, $v = e^{-i\theta}$, and $x = -(u + v)/2$. With $c = |\cos \theta|$,

$$|(x - u)^3(u + x)^2(u + 3x)^3| = (1 + 3c^2)^3(1 - c^2) \leq 9,$$

the maximum of the last expression occurring at $c^2 = 2/3$. Also

$$1 + |x| + |u^2 + 2ux + 2x^2| \leq 3.$$

Consequently $\|r\|_{H^\infty(\mathbb{D})} \leq 9/16 < 1$. The Nevanlinna–Pick criterion, with the usual Carathéodory–Schur limiting interpretation at the double node, gives the reduced Pick certificate on the double-root stratum.

If three roots coalesce, the centred spectrum is $\{x, x, x, -3x\}$. In this case $a_2 = -6x^2$, $a_1 = 8x^3$, $a_0 = -3x^4$, and substitution gives

$$r(L) \equiv 0.$$

Thus the reduced Pick matrix is exactly the Szego kernel and is positive semidefinite. Higher confluent cases are contained in this triple-root case. $\square$

Proposition 3.17 (Radial square interior model). *Let*

$$\lambda = \rho\{1, i, -1, -i\}, \quad 0 \leq \rho \leq 1.$$

Then the reduced values are

$$r = (-\rho^{10}, \rho^{10}, -\rho^{10}, \rho^{10}),$$

and every normalized principal determinant

$$\frac{\det \Pi_\lambda[I]}{\det K_\lambda[I]}, \quad K_\lambda = [(1 - \lambda_i \overline{\lambda_j})^{-1}],$$

is non-negative. The full 4×4 determinant ratio is

$$\begin{aligned} & \frac{\det \Pi_\lambda}{\det K_\lambda} \\ &= (1 - \rho)^4(1 + \rho)^4(1 + \rho^2)^4(1 + \rho^4)^4(1 + \rho^8)^2 \\ & \quad \times (\rho^2 - \rho + 1)^2(\rho^2 + \rho + 1)^2(\rho^4 - \rho^2 + 1)^2(\rho^8 - \rho^4 + 1)^2. \end{aligned}$$

Thus the radial square model is strictly positive for $0 \leq \rho < 1$ and degenerates only at the sharp square boundary point $\rho = 1$.

Proof. For this spectrum $a_2 = a_1 = 0$, $a_0 = -\rho^4$, hence

$$r(z) = 768^{-1}(c_1 + c_2 z + c_3 z^2) = -\rho^8 z^2.$$

Evaluating at $z = \rho, i\rho, -\rho, -i\rho$ gives the displayed reduced values. The determinant ratios then diagonalise in the Fourier basis of the cyclic square; direct multiplication gives the displayed factorisation. All factors are non-negative on $0 \leq \rho \leq 1$, and only the factor $(1 - \rho)^4$ vanishes in this interval. $\square$

Remark 3.18 (Final normalized determinant certificate). *The non-normal proof is reduced to the following finite interior statement. For every centred, non-confluent quadruple $\lambda_j \in \mathbb{D}$, every principal normalized determinant*

$$\mathcal{D}_I(\lambda) := \frac{\det \Pi_\lambda[I]}{\det K_\lambda[I]}, \quad \emptyset \neq I \subset \{1, 2, 3, 4\},$$

is non-negative. Moreover, the only zero sequences in the compact closure are those already covered by the boundary, confluent, or radial-square models.

Equivalently, after fixing an ordering of the four nodes and applying the Schur algorithm to the reduced data, let $\gamma_1, \dots, \gamma_4$ be the successive Schur parameters. Then

$$\frac{\det \Pi_\lambda}{\det K_\lambda} = \prod_{m=1}^4 (1 - |\gamma_m|^2)^{5-m}.$$

Thus the remaining certificate may be stated without determinants: prove

$$|\gamma_m| \leq 1, \quad m = 1, \dots, 4,$$

on the genuinely interior, non-confluent cells, with equality only on the already isolated boundary or radial-square faces. The first inequality is Proposition 3.11; the two-angle 3+1 and confluent faces are covered by Propositions 3.14 and 3.16.

The denominator $\det K_\lambda[I]$ is strictly positive for distinct interior nodes. Boundary strata are covered by Proposition 3.14 and Proposition 3.10; confluent strata are covered by Proposition 3.16; and the only remaining interior degeneration compatible with equality in the certified models is the radial square family of Proposition 3.17. Hence a proof of the displayed polynomial inequalities for $\mathcal{D}_I$ on the remaining compact semialgebraic interior cells is equivalent to the global reduced Pick criterion. In practice one clears the positive Szego determinant denominator, uses the centre constraint $\lambda_4 = -(\lambda_1 + \lambda_2 + \lambda_3)$, and certifies the finitely many real polynomial numerators by Bernstein subdivision away from the already-listed boundary faces.

Proposition 3.19 (Finite interior reduction). *Fix an ordering of the four nodes and write $\lambda_j = x_j + iy_j$, $u_j = 1 - x_j^2 - y_j^2$, with $\lambda_4 = -(\lambda_1 + \lambda_2 + \lambda_3)$. On the open cell*

$$u_j > 0, \quad \lambda_i \neq \lambda_j \quad (i \neq j),$$

the global non-normal $n = 4$ sharp bound follows from the finite family of real polynomial inequalities

$$P_{I,\alpha}(x_1, y_1, x_2, y_2, x_3, y_3) \geq 0, \quad \emptyset \neq I \subset \{1, 2, 3, 4\}, \quad \alpha \in \mathfrak{S}_4, \quad (3.11)$$

where $P_{I,\alpha}$ is obtained as follows:

1. *order the nodes by α and form the reduced Pick matrix*

$$\Pi_{ij}^{\text{red}} = \frac{1 - r_i \bar{r}_j}{1 - \lambda_i \bar{\lambda}_j}, \quad r_i = 768^{-1}(c_1 + c_2 \lambda_i + c_3 \lambda_i^2);$$

2. *take the principal minor $\det \Pi^{\text{red}}[I]$;*

3. *multiply by the positive denominator*

$$\prod_{i,j \in I} (1 - \lambda_i \bar{\lambda}_j) \prod_{1 \leq p < q \leq 4} |\lambda_p - \lambda_q|^{2e_{I,\alpha}},$$

where $e_{I,\alpha}$ is the minimal exponent clearing the Schur-complement denominators for the chosen ordering;

4. replace y_4, x_4 by the centred expressions and clear the remaining power of 768.

The polynomials $P_{I,\alpha}$ are invariant under common rotation and under simultaneous conjugation. Hence one may fix $y_1 = 0$, split into the six real order chambers

$$|\lambda_{\alpha(1)}| \geq |\lambda_{\alpha(2)}| \geq |\lambda_{\alpha(3)}| \geq |\lambda_{\alpha(4)}|,$$

and verify (3.11) by Bernstein subdivision on each chamber after adjoining the chamber inequalities

$$u_j \geq 0, \quad |\lambda_{\alpha(m)}|^2 - |\lambda_{\alpha(m+1)}|^2 \geq 0.$$

All boundary faces of this compactification are already covered: $u_j = 0$ by the normal and 3+1 boundary certificates, $\lambda_i = \lambda_j$ by Proposition 3.16, and the radial square degeneration by Proposition 3.17. Consequently a Bernstein certificate for the finitely many chamber polynomials $P_{I,\alpha}$ closes the non-normal case without any further analytic argument.

Proof. For distinct interior nodes the Szego matrix

$$K_{ij} = (1 - \lambda_i \overline{\lambda_j})^{-1}$$

is strictly positive definite. Sylvester's criterion applied to the reduced Pick matrix is therefore equivalent to non-negativity of all normalized principal determinants

$$\det \Pi^{\text{red}}[I] / \det K[I].$$

The displayed clearing operation multiplies each normalized determinant by a strictly positive factor on the open cell, so it does not change the sign. The denominators introduced by Schur complements are powers of $1 - \lambda_i \overline{\lambda_j}$ and of the squared Vandermonde factors; these are positive away from boundary and collision faces. After the centre constraint is imposed, real and imaginary parts are ordinary real polynomials. Rotation and conjugation invariance give the stated dimension reduction and allow the chamber split. Finally, every non-compact way of approaching the closure is one of the listed faces, and those faces have already been handled by the preceding propositions. Thus the finite Bernstein verification on the remaining compact chambers is equivalent to the desired global reduced Pick positivity. $\square$

Proposition 3.20 (Analytic certificate for the interior reduced Pick criterion). *The reduced Pick matrix $\Pi_{\lambda}^{\text{red}}$ is positive semidefinite for every centred, non-confluent quadruple $\lambda_1, \lambda_2, \lambda_3, \lambda_4 \in \overline{\mathbb{D}}$.*

Proof. Propositions 3.14, 3.16, and 3.17, together with Proposition 3.10 and Lemma 3.13, reduce the claim to interior non-confluent quadruples with $\max_j |\lambda_j| < 1$. By Proposition 3.19, the remaining verification is a finite list of real polynomial non-negativity statements on rational boxes. All non-corner boxes are certified by the Bernstein tables described in Appendix B. The only box on which the direct Bernstein expansion has negative coefficients is the third Schur-pivot hard corner in the two-parameter 3+1 chart,

$$\rho \in [3/4, 1], \quad y \in [3/4, 1],$$

near the square endpoint $(\rho, y) = (1, 1)$.

Appendix E resolves that corner by the blow-up

$$u = 1 - \rho, \quad v = 1 - y, \quad u = 4v + vz.$$

In these coordinates the primitive numerator $Q(u, v)$ of $1 - |\gamma_3|^2$ has lowest face

$$C v^9 (u - 4v)^2, \quad C = 382511685112441262309376 > 0.$$

Thus the direct Bernstein failure is caused by a square ridge, not by a sign change. The resolved polynomial

$$S(z, v) = Q(4v + vz, v)/v^{11}$$

is positive in four regions: the inner tube $|z| \leq 16v$, the annulus $16v < |z| < 1/4$, the fixed- z range $1/4 \leq |z| \leq 32$, and the positive tail $z \geq 32$ after the reciprocal chart $v = ru$. The first and second regions are closed by explicit coefficient and remainder inequalities, and the latter two by exact rational Bernstein expansions with no negative coefficients.

Consequently every Schur pivot polynomial in the finite interior reduction is non-negative, with zeros only on the already certified boundary or confluent faces. On the non-confluent interior the pivots are strictly positive, hence the Schur recursion gives $\Pi_\lambda^{\text{red}} \succ 0$. Taking limits gives positive semidefiniteness on the closed non-confluent locus. $\square$

Since the reduced Pick criterion (3.10) holds globally by Proposition 3.20,

$$\|A_B\|_2 = 768 \|q_\lambda(S_B)\|_2 \leq 768,$$

and therefore $\|A_B\|_* \leq 4\|A_B\|_2 \leq 3072$.

In summary, Conjecture 3.6 for $n = 4$ is established on the normal strata and on the paired, 3+1, and radial two-pair Blaschke boundary strata listed in Proposition 3.10. The first reduced pivot is established on the full centred disk by Proposition 3.11, and the confluent rank-one defect layers are covered by Proposition 3.16. The radial square interior degeneration is isolated in Proposition 3.17. The final step—positive semidefiniteness of the reduced Pick matrix on the genuinely interior, non-confluent cells—is established by the finite Bernstein and blow-up certificate in Proposition 3.20.

Remark 3.21 (Auxiliary coefficient region). One elementary ingredient behind Theorem 3.1 is the coefficient-region inequality

$$|q| + \frac{2}{9}|p|^2 \leq 1$$

for centred cubic contractions. A short Schur-Cohn derivation is included in Appendix F; Appendix G then uses it with von Neumann's inequality for contractions to obtain the exact operator-norm constant.

3.1 A universal coefficient-box constant

The sharp cubic constant will be used in the examples, but the graph argument only needs some Lipschitz control of the discriminant. For arbitrary matrix size one can obtain such a bound from the coefficient expression of the discriminant, as follows.

For a monic degree- n polynomial

$$P(E) = E^n + a_{n-1}E^{n-1} + \cdots + a_0,$$

write $\mathcal{D}_n(a_0, \dots, a_{n-1})$ for its classical discriminant. For $0 \leq j \leq n-1$, set

$$A_j(R) = \binom{n}{n-j} R^{n-j}, \quad F_{n,j}(R) = \binom{n}{n-j} (n-j) R^{n-j-1},$$

and

$$C_{n,j}(R) = \sup_{|a_\ell| \leq A_\ell(R)} \left| \frac{\partial \mathcal{D}_n}{\partial a_j}(a_0, \dots, a_{n-1}) \right|.$$

Define

$$\mathcal{G}_n^{\text{box}}(R) = \sum_{j=0}^{n-1} C_{n,j}(R) F_{n,j}(R).$$

Proposition 3.22 (Universal discriminant Lipschitz constant). *For all $A, B \in M_n(\mathbb{C})$ with $\|A\|_2, \|B\|_2 \leq R$,*

$$|\text{Disc}(A) - \text{Disc}(B)| \leq \mathcal{G}_n^{\text{box}}(R) \|A - B\|_2.$$

The same formula, with the box restricted by linear coefficient constraints such as $a_{n-1} = 0$, gives a valid structured constant on the corresponding convex matrix class.

Proof. Write $\det(EI - A) = E^n + \sum_{j=0}^{n-1} a_j(A) E^j$. The coefficient $a_j(A)$ is, up to sign, the trace of the $(n-j)$ -th exterior power $\bigwedge^{n-j} A$; equivalently it is the sum of the $(n-j) \times (n-j)$ principal minors of A . With $m = n - j$, the exterior-power Leibniz estimate gives

$$\left\| \bigwedge^m A - \bigwedge^m B \right\|_2 \leq m R^{m-1} \|A - B\|_2,$$

because every term in the telescoping expansion contains one factor $A - B$ and $m - 1$ factors bounded by R . Since $\dim \bigwedge^m \mathbb{C}^n = \binom{n}{m}$,

$$|a_j(A) - a_j(B)| \leq \binom{n}{m} m R^{m-1} \|A - B\|_2 = F_{n,j}(R) \|A - B\|_2.$$

Also $|a_j(A)| \leq A_j(R)$, and similarly for B . Hence the straight segment between the two coefficient vectors lies in the coefficient box $|a_j| \leq A_j(R)$. Applying the fundamental theorem of calculus to $\mathcal{D}_n$ along this segment gives

$$|\mathcal{D}_n(a(A)) - \mathcal{D}_n(a(B))| \leq \sum_{j=0}^{n-1} C_{n,j}(R) |a_j(A) - a_j(B)|,$$

which is the stated estimate. If a convex structural class imposes linear coefficient constraints, the same proof is performed on the smaller coefficient region. $\square$

Remark 3.23 (Role of the general bound). *Proposition 3.22 is not intended to be sharp; it is the finite-dimensional fallback that makes the graph certificate valid for all n . The sharp cubic constant in Theorem 3.1 shows what is gained when the structure of a particular matrix class is exploited. In higher dimension the constants inherit the high degree of the resultant and the dimension factor in inequalities such as $\|A\|_* \leq n\|A\|_2$, so structured constants become increasingly important.*

4 A graph certificate for braid monodromy representations

The whole complement X_H may be hard to represent computationally. In practice one chooses a compact graph $G \subset X_H$ carrying the loops of interest. For a punctured plane this can be a lollipop or spider graph whose loops generate the free group of meridians. For a sampled domain it can be a subgraph of a triangulation avoiding the discriminant locus.

Let G be a compact finite graph and let

$$H : G \longrightarrow M_n(\mathbb{C})$$

be continuous with simple spectrum on G . Define the graph discriminant margin

$$M_G(H) = \inf_{x \in G} |\text{Disc}(H(x))|.$$

If $M_G(H) > 0$, then H induces a representation

$$\beta_{H,G} : \pi_1(G, x_0) \longrightarrow B_n.$$

When $G \hookrightarrow X_H$ carries specified generators of $\pi_1(X_H, p_0)$, this representation gives the corresponding part of the full spectral braid monodromy.

The stability estimate is phrased in terms of the following Lipschitz constant.

Definition 4.1 (Matrix-space discriminant Lipschitz constant). *Let $\mathcal{S} \subset M_n(\mathbb{C})$ be a convex matrix class, for instance all matrices, trace-zero matrices, or a structured affine subspace. A number $\mathcal{G}_{\mathcal{S}}(R)$ is called an admissible discriminant Lipschitz constant on $\mathcal{S} \cap \{\|A\|_2 \leq R\}$ if*

$$|\text{Disc}(A) - \text{Disc}(B)| \leq \mathcal{G}_{\mathcal{S}}(R) \|A - B\|_2$$

for all $A, B \in \mathcal{S}$ with $\|A\|_2, \|B\|_2 \leq R$.

For the full matrix ball, Proposition 3.22 supplies a coefficient-box bound. In the trace-zero cubic case, Theorem 3.1 gives the sharp value $\mathcal{G}_{\text{tr}=0,3}(R) = 162R^5$. The slightly larger radius $R_* > R$ allows for the perturbed family: the condition $\rho < R_* - R$ keeps the straight-line homotopy inside the ball on which the chosen Lipschitz constant is valid.

Theorem 4.2 (Graph stability of braid monodromy). *Let $\mathcal{S} \subset M_n(\mathbb{C})$ be convex. Let $H, \tilde{H} : G \rightarrow \mathcal{S}$ be continuous maps on a compact graph G . Suppose*

$$\sup_{x \in G} \|H(x)\|_2 \leq R, \quad M_G(H) = \inf_{x \in G} |\text{Disc}(H(x))| > 0.$$

Choose $R_* > R$ and assume

$$\sup_{x \in G} \|\tilde{H}(x) - H(x)\|_2 < \rho, \quad \rho < R_* - R,$$

and

$$\mathcal{G}_{\mathcal{S}}(R_*)\rho < \frac{M_G(H)}{2}.$$

Then the straight-line homotopy

$$H_t = (1-t)H + t\tilde{H}, \quad 0 \leq t \leq 1,$$

stays in the discriminant complement over G . Hence

$$\beta_{H,G} = \beta_{\tilde{H},G}$$

up to simultaneous conjugacy.

Proof. The norm bound gives $\|H_t(x)\|_2 < R_*$ for all $x \in G$ and $t \in [0, 1]$. By the admissible Lipschitz estimate,

$$\begin{aligned} |\text{Disc}(H_t(x)) - \text{Disc}(H(x))| &\leq \mathcal{G}_{\mathcal{S}}(R_*) \|H_t(x) - H(x)\|_2 \\ &\leq \mathcal{G}_{\mathcal{S}}(R_*)\rho < \frac{M_G(H)}{2}. \end{aligned}$$

Therefore $|\text{Disc}(H_t(x))| \geq M_G(H)/2 > 0$. The homotopy invariance lemma applies. $\square$

Corollary 4.3 (Certified generator list). *Let $\gamma_1, \dots, \gamma_m$ be based loops in G representing a chosen finite list of generators. Under the hypotheses of Theorem 4.2, the braid words computed from $H \circ \gamma_j$ and $\tilde{H} \circ \gamma_j$ agree for all j , after the same basepoint conjugacy.*

Corollary 4.4 (Structured perturbation radius). *Let $\mathcal{L}$ be a linear subspace of admissible perturbation directions inside the convex class $\mathcal{S}$, for example trace-preserving, Hamiltonian, or a fixed anti-linear symmetry class. If $\mathcal{G}_{\mathcal{L}}(R_*)$ is an admissible Lipschitz constant for Disc restricted to line segments in directions from $\mathcal{L}$, then Theorem 4.2 remains true with $\mathcal{G}_{\mathcal{S}}(R_*)$ replaced by $\mathcal{G}_{\mathcal{L}}(R_*)$. In particular the certified radius is multiplied by $\mathcal{G}_{\mathcal{S}}(R_*)/\mathcal{G}_{\mathcal{L}}(R_*)$ whenever the restricted constant is smaller. For example, for trace-zero cubic perturbations one may take $\mathcal{G}_{\mathcal{L}}(R_*) = 162R_*^5$ by Theorem 3.1. In the companion cubic section below this is precisely the structured constant used in place of the much larger unstructured coefficient-box constant.*

Proof. The proof of Theorem 4.2 only uses a Lipschitz bound for Disc along the straight-line homotopy between H and $\tilde{H}$. If all perturbation directions lie in $\mathcal{L}$, that homotopy stays in the corresponding restricted class, so the restricted constant applies. $\square$

Corollary 4.5 (Certified relations and Zariski–van Kampen consistency). *Let G be a based graph and let $g_1, \dots, g_m \in \pi_1(G, x_0)$ be loops satisfying a word relation $w(g_1, \dots, g_m) = 1$ in $\pi_1(G, x_0)$. Suppose the hypotheses of Theorem 4.2 hold and write $b_j = \beta_{H,G}(g_j) \in B_n$. Then*

$$w(b_1, \dots, b_m) = 1 \quad \text{in } B_n,$$

and the same relation holds for every perturbation inside the certified radius. In particular, for a punctured Riemann sphere with meridians $g_1, \dots, g_m, g_\infty$ satisfying $g_1 \cdots g_m g_\infty = 1$, the certified braids satisfy

$$b_1 \cdots b_m b_\infty = 1.$$

Equivalently, on a large affine circle the braid is the inverse of the positive meridian at infinity, with the convention fixed by the local coordinate $w = 1/k$.

Proof. The braid assignment is the induced homomorphism $\beta_{H,G} : \pi_1(G, x_0) \rightarrow B_n$, so it sends every relation in $\pi_1(G, x_0)$ to the identity in B_n . The perturbation statement is Theorem 4.2. The displayed sphere relation is the standard Zariski–van Kampen relation for meridians of a punctured sphere. $\square$

Remark 4.6 (Why a graph). *The graph formulation separates topology from numerics. Topology enters by choosing a graph that carries the desired fundamental-group information. Numerics enters by certifying that the eigenvalue map is well defined and stable on that graph. This is weaker than certifying the whole complement, but it is exactly what a finite braid-monodromy computation uses.*

5 Finite sampling on graph edges

The graph margin $M_G(H)$ is still an infimum over a continuum. This section gives a finite-sampling replacement.

Let every edge e of G be parametrised by $\gamma_e : [0, 1] \rightarrow G$. Assume that $H \circ \gamma_e$ is Lipschitz in operator norm:

$$\|H(\gamma_e(s)) - H(\gamma_e(t))\|_2 \leq L_e |s - t|, \quad s, t \in [0, 1].$$

Set

$$L_G = \max_e L_e.$$

Let $S_e \subset [0, 1]$ be a finite sample set on edge e , and define the mesh radius

$$h_G = \max_e \sup_{t \in [0, 1]} \min_{s \in S_e} |t - s|.$$

For uniform N -point sampling on a closed loop parametrised by the angular variable $\theta \in [0, 2\pi]$, one may take $h_G = \pi/N$ in that angular parametrisation. If the same loop is reparametrised by $t = \theta/(2\pi) \in [0, 1]$, the corresponding mesh radius is $1/(2N)$ and the Lipschitz constant must be rescaled accordingly.

Define the sampled discriminant margin

$$\widehat{M}_G = \min_e \min_{s \in S_e} |\text{Disc}(H(\gamma_e(s)))|.$$

Proposition 5.1 (Sampling correction). *Assume $\|H(x)\|_2 \leq R$ on G , choose $R_* \geq R$, and let $\mathcal{G}_S(R_*)$ be admissible on the matrix class containing the image of H . Then*

$$M_G(H) \geq \widehat{M}_G - \mathcal{G}_S(R_*) L_G h_G.$$

In particular, if

$$M_G^{\text{cert}} := \widehat{M}_G - \mathcal{G}_S(R_*) L_G h_G > 0,$$

then H has simple spectrum on all of G .

Proof. For any $x = \gamma_e(t)$ choose a sample $s \in S_e$ with $|t - s| \leq h_G$. The Lipschitz bound gives

$$\|H(\gamma_e(t)) - H(\gamma_e(s))\|_2 \leq L_G h_G.$$

Applying the discriminant Lipschitz estimate yields

$$|\text{Disc}(H(\gamma_e(t))) - \text{Disc}(H(\gamma_e(s)))| \leq \mathcal{G}_S(R_*) L_G h_G.$$

Taking the minimum over all $x \in G$ gives the claim. $\square$

Theorem 5.2 (Sampled certified braid representation). *Let $H : G \rightarrow \mathcal{S}$ satisfy the hypotheses above and suppose $M_G^{\text{cert}} > 0$. Let $\tilde{H} : G \rightarrow \mathcal{S}$ be another family with*

$$\sup_{x \in G} \left\| \tilde{H}(x) - H(x) \right\|_2 < \rho, \quad \rho < R_* - R,$$

and

$$\mathcal{G}_S(R_*) \rho < \frac{M_G^{\text{cert}}}{2}.$$

Then H and $\tilde{H}$ induce the same braid monodromy representation on $\pi_1(G)$, up to simultaneous conjugacy.

Proof. Proposition 5.1 gives $M_G(H) \geq M_G^{\text{cert}} > 0$. Theorem 4.2 applies with this certified lower bound. $\square$

Remark 5.3 (One-line operational form). *A completely sampled certificate is therefore*

$$\rho < \frac{\widehat{M}_G - \mathcal{G}_S(R_*) L_G h_G}{2 \mathcal{G}_S(R_*)},$$

provided the numerator is positive. This is the direct multi-loop analogue of the single-loop correction $\widehat{M} - \mathcal{G}_S(R_*) L_H \pi/N$ when the loop is parametrised by the angular variable.

Proposition 5.4 (Sampling complexity for a target radius). *In the notation of Proposition 5.1, suppose one wants to certify every perturbation of size at least $\rho_{\text{tar}} > 0$. It is sufficient to choose the mesh radius so that*

$$h_G < \frac{\widehat{M}_G - 2\rho_{\text{tar}}\mathcal{G}_S(R_*)}{\mathcal{G}_S(R_*)L_G},$$

with positive numerator. If an edge e is parametrised by arclength and has length ℓ_e , then a uniform grid with

$$N_e \geq \left\lceil \frac{\ell_e}{2h_G} \right\rceil$$

subintervals on that edge is enough.

Proof. The sampled certificate gives radius

$$\frac{\widehat{M}_G - \mathcal{G}_S(R_*)L_G h_G}{2\mathcal{G}_S(R_*)}.$$

Requiring this to exceed ρ_{tar} gives the displayed inequality. For a uniform arclength mesh with subinterval size ℓ_e/N_e , every point lies within $\ell_e/(2N_e)$ of a sample. $\square$

Proposition 5.5 (Algorithmic certificate). *For computations, the certificate reduces to the following finite procedure on a chosen graph G .*

- (i) Choose based graph loops $\gamma_1, \dots, \gamma_m$ representing the required generators.
- (ii) Sample each graph edge and compute $\widehat{M}_G$.
- (iii) Bound R_* , L_G , and choose an admissible constant $\mathcal{G}_S(R_*)$.
- (iv) Verify $M_G^{\text{cert}} > 0$.
- (v) Track the eigenvalues along the generator loops to obtain braid words $b_1, \dots, b_m \in B_n$.

If a perturbation satisfies the radius in Theorem 5.2, then the same generator list $(b_1, \dots, b_m)$ is certified for the perturbed family.

6 Adaptive graph certificates from eigenvalue gaps

A discriminant margin is convenient because it is polynomial in the matrix entries. Away from the discriminant it may nevertheless underestimate the available separation. On graph regions where the matrices are uniformly diagonalizable, one can certify the same homotopy using eigenvalue gaps directly, via Bauer–Fike.

For a diagonalizable matrix $A = V\Lambda V^{-1}$, let

$$\kappa(A) = \inf_{A=V\Lambda V^{-1}} \|V\|_2 \|V^{-1}\|_2,$$

where the infimum is over diagonalizations. For a graph family define

$$\delta_G(H) = \inf_{x \in G} \min_{i < j} |\lambda_i(H(x)) - \lambda_j(H(x))|, \quad K_G(H) = \sup_{x \in G} \kappa(H(x)).$$

Proposition 6.1 (Bauer–Fike graph certificate). *Let $H : G \rightarrow M_n(\mathbb{C})$ be continuous, diagonalizable on G , and suppose $\delta_G(H) > 0$, $K_G(H) < \infty$. If $\tilde{H} : G \rightarrow M_n(\mathbb{C})$ satisfies*

$$\sup_{x \in G} \left\| \tilde{H}(x) - H(x) \right\|_2 < \frac{\delta_G(H)}{2K_G(H)},$$

then the straight-line homotopy from H to $\tilde{H}$ has simple spectrum on G . Consequently H and $\tilde{H}$ induce the same braid monodromy representation on $\pi_1(G)$, up to simultaneous conjugacy.

Proof. For each fixed x , the Bauer–Fike theorem [14] places every eigenvalue of $H(x) + E$ inside the union of disks of radius $K_G(H) \|E\|_2$ centred at the eigenvalues of $H(x)$. Along the straight-line homotopy $E = t(\tilde{H}(x) - H(x))$, this radius is strictly less than $\delta_G(H)/2$. The disks around distinct eigenvalues of $H(x)$ are therefore disjoint, so no collision can occur. Homotopy invariance in configuration space gives the braid statement. $\square$

Proposition 6.2 (Sampled adaptive gap correction). *Assume in addition that H is L_G -Lipschitz on graph edges and that $K_G(H) \leq K$ on the whole graph, not merely at the sampled points. Let*

$$\hat{\delta}_G = \min_e \min_{s \in S_e} \min_{i < j} |\lambda_i(H(\gamma_e(s))) - \lambda_j(H(\gamma_e(s)))|.$$

Then

$$\delta_G(H) \geq \hat{\delta}_G - 2KL_G h_G.$$

In particular, if $\delta_G^{\text{cert}} := \hat{\delta}_G - 2KL_G h_G > 0$, then every perturbation satisfying

$$\sup_{x \in G} \left\| \tilde{H}(x) - H(x) \right\|_2 < \frac{\delta_G^{\text{cert}}}{2K}$$

preserves the braid monodromy representation on $\pi_1(G)$.

Proof. Let x and y lie on the same edge with parameter distance at most h_G , with y a sampled point. Then $\|H(x) - H(y)\|_2 \leq L_G h_G$. If $\hat{\delta}_G - 2KL_G h_G \leq 0$, the displayed lower bound is trivial, since $\delta_G(H) \geq 0$. Assume therefore that $\hat{\delta}_G - 2KL_G h_G > 0$. Then the disks of radius $KL_G h_G$ centred at the distinct eigenvalues of $H(y)$ are disjoint, because their centres are separated by at least $\hat{\delta}_G$. Bauer–Fike places every eigenvalue of $H(x)$ in the union of these disks. Since there are exactly n eigenvalues counted with multiplicity and the disks are disjoint, each disk contains exactly one eigenvalue of $H(x)$. This gives a bijective matching between the spectra of $H(y)$ and $H(x)$, and each matched pair is within $KL_G h_G$. Hence every pairwise gap decreases by at most $2KL_G h_G$. Taking the minimum over sampled points and graph points proves the estimate. The final statement is Proposition 6.1 with the certified lower bound. $\square$

Lemma 6.3 (Propagating diagonalizer condition numbers). *Let an edge be parametrised by $t \in [0, 1]$, and suppose on this edge a continuous diagonalization $H(t) = V(t)\Lambda(t)V(t)^{-1}$ has been enclosed with*

$$\|V(t) - V(s)\|_2 \leq L_V |t - s|, \quad \|V(t)^{-1} - V(s)^{-1}\|_2 \leq L_W |t - s|.$$

For a sample set of mesh radius h , define

$$\hat{V} = \max_{s \in S} \|V(s)\|_2, \quad \hat{W} = \max_{s \in S} \|V(s)^{-1}\|_2.$$

Then on the whole edge

$$\kappa(H(t)) \leq (\widehat{V} + L_V h)(\widehat{W} + L_W h).$$

Consequently Proposition 6.2 becomes a fully sampled certificate whenever interval arithmetic or a continuation method supplies L_V and L_W .

Proof. Choose a sample s with $|t - s| \leq h$. Then $\|V(t)\|_2 \leq \|V(s)\|_2 + L_V h$ and similarly for the inverse. Multiplying the two bounds gives the estimate. $\square$

Remark 6.4 (Certifying the condition-number bound). *The adaptive certificate is complementary to the discriminant certificate. It is often much larger on normal or well-conditioned portions of the graph, where $K \approx 1$. The hypothesis $K_G(H) \leq K$, however, is a uniform condition on the whole graph, not a sample-only assertion. A reproducible implementation must therefore propagate the sampled condition numbers along each edge, for instance by enclosing a diagonalizer and its inverse on small edge intervals or by using interval arithmetic for the eigenprojectors. If no such enclosure is available, Proposition 5.1 remains the safer fully polynomial certificate, because it uses only the matrix-entry Lipschitz bound and the discriminant Lipschitz constant. The adaptive certificate becomes weak near exceptional points, precisely where diagonalizability degenerates and the discriminant formulation is more natural.*

7 A non-abelian cubic arrangement

In the planar examples the discriminant locus is a finite set and the parameter complement is a punctured plane. We use a lollipop graph: disjoint base arcs from a common basepoint to marked points on small positive meridian circles around the punctures, with each circle attached to the basepoint by its base arc. The certificate is evaluated on the whole graph, including the base arcs. The cyclic cubic example below has a transparent global factorisation but cyclic image; the following companion family is the minimal cubic example in which different meridians produce different adjacent half-twists.

Consider the monic polynomial

$$P_k(E) = E^3 - 3E - k, \quad k \in \mathbb{C},$$

and its companion matrix

$$C(k) = \begin{pmatrix} 0 & 0 & k \\ 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix}.$$

Then $\det(EI - C(k)) = P_k(E)$, $\operatorname{tr} C(k) = 0$, and

$$\Delta(k) = \operatorname{Disc}(P_k) = 108 - 27k^2 = 27(4 - k^2).$$

Thus the discriminant complement is

$$X = \mathbb{C} \setminus \{-2, 2\}.$$

The two punctures are the critical values of the cubic map

$$f(E) = E^3 - 3E, \quad f'(E) = 3(E^2 - 1).$$

Indeed $f(-1) = 2$ and $f(1) = -2$. At $k = 2$ the double root is $E = -1$, while at $k = -2$ the double root is $E = 1$.

Proposition 7.1 (Two-puncture cubic braid monodromy). *Let $C(k)$ be the companion family above. Take the basepoint $\kappa_0 = 0$ and order the three real roots of $P_0(E) = E(E^2 - 3)$ as*

$$-\sqrt{3} < 0 < \sqrt{3}.$$

With Convention 2.3, a positive meridian around 2 gives the Artin generator σ_1 , and a positive meridian around -2 gives σ_2 , up to the simultaneous conjugacy associated with the chosen base paths. Consequently the spectral braid monodromy representation

$$\beta_C : \pi_1(\mathbb{C} \setminus \{-2, 2\}, 0) \longrightarrow B_3$$

has non-commuting generator images.

Proof. The discriminant computation shows that the only branch values are ± 2 . Near $k = 2$, write $E = -1 + u$ and $k = 2 + s$. Since

$$(-1 + u)^3 - 3(-1 + u) - (2 + s) = 3u^2 + u^3 - s,$$

the two local roots satisfy $u = \pm(s/3)^{1/2} + O(s)$. Along the real base arc from 0 to $2 - r$, the roots remain real and ordered. The branch tracking is elementary: if $P_k(E) = 0$ and $E \neq \pm 1$, then

$$\frac{dE}{dk} = \frac{1}{3(E^2 - 1)}.$$

At the basepoint, the derivative is positive on the branch starting at $-\sqrt{3}$, negative on the branch starting at 0, and positive on the branch starting at $\sqrt{3}$. Hence, as k increases from 0 to 2, the first branch increases toward -1 , the middle branch decreases toward -1 , and the largest branch increases toward the simple root 2. The limiting roots at $k = 2$ are fixed by $P_2(E) = (E - 2)(E + 1)^2$, so the colliding pair, measured from the basepoint ordering $-\sqrt{3} < 0 < \sqrt{3}$, occupies positions 1 and 2. With Convention 2.3, $s^{1/2}$ gains a factor -1 after one positive circuit and the local pair makes the positive half-twist of positions 1 and 2. This is σ_1 :

puncture	colliding roots at the basepoint	braid
$k = 2$	$(-\sqrt{3}, 0)$	σ_1

Near $k = -2$, write $E = 1 + u$ and $k = -2 + s$. Then

$$(1 + u)^3 - 3(1 + u) - (-2 + s) = 3u^2 + u^3 - s,$$

so the two local roots near $E = 1$ are exchanged by the same square-root monodromy. Along the real base arc from 0 to $-2 + r$, the same derivative formula, now with k decreasing, shows that the branch starting at $-\sqrt{3}$ decreases to the simple root -2 , while the branches starting at 0 and $\sqrt{3}$ move toward the double root 1, as is confirmed by $P_{-2}(E) = (E + 2)(E - 1)^2$. Thus the colliding pair occupies positions 2 and 3, and the same Convention 2.3 gives the positive half-twist of those positions:

puncture	colliding roots at the basepoint	braid
$k = -2$	$(0, \sqrt{3})$	σ_2

The two Artin generators do not commute in B_3 . Different choices of base arcs or projection convention conjugate all images simultaneously or invert the meridian orientation, but do not change the fact that the two local meridians correspond to adjacent non-commuting half-twists. $\square$

Remark 7.2 (Why this example is useful). *The cyclic $E^3 = w$ example has a three-cycle around every finite puncture. By contrast, the family $E^3 - 3E = k$ separates the two simple critical values of a cubic map. It is the finite-dimensional spectral analogue of the elementary braid-monodromy factorisation of a degree-three polynomial map into two adjacent half-twists.*

7.1 Image, kernel, and the braid at infinity

The two local half-twists determine the whole image of the representation. Let

$$F_2 = \langle g_+, g_- \rangle$$

where g_+ and g_- are the positive meridians around 2 and -2 , respectively, with the convention of Proposition 7.1.

Proposition 7.3 (Image and kernel of the two-puncture cubic monodromy). *For the companion cubic family $P_k(E) = E^3 - 3E - k$, the spectral braid monodromy representation*

$$\beta_C : F_2 \longrightarrow B_3$$

satisfies

$$\beta_C(g_+) = \sigma_1, \quad \beta_C(g_-) = \sigma_2.$$

Consequently $\text{Im } \beta_C = B_3$. Moreover β_C is not faithful: its kernel is the normal closure in F_2 of the Artin relation

$$g_+ g_- g_+ (g_- g_+ g_-)^{-1}.$$

Thus every braid in B_3 is realised by a loop in $\mathbb{C} \setminus \{-2, 2\}$, but different loops can realise the same braid.

Proof. The first two identities are Proposition 7.1. The Artin presentation

$$B_3 = \langle \sigma_1, \sigma_2 \mid \sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2 \rangle$$

then shows immediately that the image is all of B_3 . Since $F_2 = \langle g_+, g_- \rangle$ is the free group on the two meridians, the homomorphism induced by $g_+ \mapsto \sigma_1$, $g_- \mapsto \sigma_2$ is exactly the quotient map from the free presentation to the Artin presentation. Its kernel is therefore the normal closure of the displayed relation. In particular the representation is surjective but not injective. $\square$

There is also a useful consistency check at infinity. Compactify the parameter line to $\mathbb{P}^1$. If g_∞ denotes a positive meridian around infinity on the Riemann sphere, then the standard sphere relation is

$$g_+ g_- g_\infty = 1.$$

This is the simplest Zariski–van Kampen relation in the present setting.

Proposition 7.4 (Braid at infinity for the two-puncture cubic). *With the conventions above,*

$$\beta_C(g_\infty) = (\sigma_1 \sigma_2)^{-1}.$$

Equivalently, a large counterclockwise circle in the affine k -plane has braid $\sigma_1 \sigma_2$, while the positively oriented meridian around infinity on $\mathbb{P}^1$ has the inverse braid.

Proof. The sphere relation gives

$$\beta_C(g_\infty) = (\beta_C(g_+)\beta_C(g_-))^{-1} = (\sigma_1\sigma_2)^{-1}.$$

The same conclusion follows from the large-parameter asymptotics. For $|k| \gg 1$, the equation $E^3 - 3E - k = 0$ has branches

$$E_j(k) = \omega^j k^{1/3} + O(k^{-1/3}), \quad j = 0, 1, 2.$$

A counterclockwise large affine circle increases $\arg k$ by 2π , so the three branches undergo the positive cyclic braid $\sigma_1\sigma_2$ under the same convention as the local cubic-root model. A positive meridian around infinity on the sphere is the same geometric circle with the opposite orientation in the affine chart: in the local coordinate $w = 1/k$ near infinity, a counterclockwise large circle in k becomes a clockwise small circle in w . Hence the infinity meridian gives the inverse braid. $\square$

7.2 Certified radius for an explicit meridian graph

Fix $0 < r < 2$. Let G_r be the based lollipop graph with basepoint 0, consisting of the two real segments

$$[0, 2 - r], \quad [0, -2 + r],$$

and the two circles

$$k = 2 + re^{it}, \quad k = -2 + re^{it}, \quad 0 \leq t \leq 2\pi.$$

The corresponding based loops go from 0 along a segment, traverse the circle once positively, and return along the same segment. This graph carries the two standard meridians of $\mathbb{C} \setminus \{-2, 2\}$.

On the right circle,

$$4 - (2 + re^{it})^2 = -4re^{it} - r^2e^{2it},$$

so

$$\min_t |4 - (2 + re^{it})^2| = r(4 - r).$$

On the right segment $0 \leq k \leq 2 - r$,

$$4 - k^2 \geq 4 - (2 - r)^2 = r(4 - r).$$

The left circle and left segment give the same bound. Hence the full graph, including the base arcs, has the exact margin

$$M_{G_r} = 27r(4 - r).$$

For trace-zero perturbations of the companion family, Theorem 3.1 gives

$$\rho_{\text{tr}0} < \frac{27r(4 - r)}{2 \cdot 162R_*^5} = \frac{r(4 - r)}{12R_*^5}.$$

With finite sampling, replace $27r(4 - r)$ by the corrected value

$$\widehat{M}_G - 162R_*^5 L_G h_G.$$

For this companion matrix the norm bound can be sharpened explicitly. A short calculation gives

$$C(k)^*C(k) = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 3 & 0 & |k|^2 + 9 \end{pmatrix},$$

so

$$\|C(k)\|_2^2 = \frac{|k|^2 + 10 + ((|k|^2 + 8)^2 + 36)^{1/2}}{2}.$$

This expression is increasing in $|k|$. On G_r , $|k| \leq 2 + r$, and this formula can be used instead of a row-sum estimate.

Example 7.5 (A concrete non-commuting certificate). *Take $r = 1/2$. Then*

$$M_{G_r} = 27 \cdot \frac{1}{2} \cdot \frac{7}{2} = \frac{189}{4}.$$

On the whole graph $|k| \leq 5/2$. The exact formula above gives $\sup_{G_r} \|C(k)\|_2 < 4$, so one may take $R_ = 4$. The trace-zero stability radius is then*

$$\rho_{\text{tr}0} < \frac{189/4}{2 \cdot 162 \cdot 4^5} \approx 1.42 \times 10^{-4}.$$

This is about three times larger than the cruder $R_ = 5$ certificate. It certifies both generator braids σ_1 and σ_2 on the same graph. The certificate is therefore for the representation $F_2 \rightarrow B_3$, not merely for two unrelated loops.*

Remark 7.6 (Size of the certified radius). *The numerical value above is still conservative because the radius scales as R_*^{-5} , and any slack in the graph norm bound is amplified by a fifth power. Here the exact singular-value formula reduces the bound from the earlier crude choice $R_* = 5$ to $R_* = 4$. In larger examples the same role can be played by interval arithmetic on the largest singular value along the graph. The adaptive certificate of Proposition 6.1 gives a separate, often much larger, radius on well-conditioned portions of the same graph.*

Remark 7.7 (Adaptive radius on the same graph). *Away from the two punctures one may also use Proposition 6.1. On parts of G_r where the companion matrix is well conditioned, the gap radius $\delta_G/(2K_G)$ can be much larger than the global discriminant radius. The discriminant certificate is nevertheless useful because it does not require diagonalization conditioning data and remains polynomial in the matrix entries.*

8 Cyclic cubic exceptional-point arrangement

Let $\omega = e^{2\pi i/3}$ and consider

$$H_3(k) = \begin{pmatrix} k & 1 & 0 \\ 0 & \omega k & 1 \\ 1 & 0 & \omega^2 k \end{pmatrix}, \quad k \in \mathbb{C}.$$

The trace is zero. A direct determinant calculation gives

$$P_k(E) = \det(EI - H_3(k)) = E^3 - k^3 - 1.$$

For a depressed cubic $E^3 + a_0$, the discriminant is $-27a_0^2$, hence

$$\Delta(k) = \text{Disc}(P_k) = -27(k^3 + 1)^2.$$

The exceptional points are

$$a_0 = -1, \quad a_1 = -\omega, \quad a_2 = -\omega^2.$$

Thus the parameter complement is the three-punctured plane

$$X = \mathbb{C} \setminus \{-1, -\omega, -\omega^2\}.$$

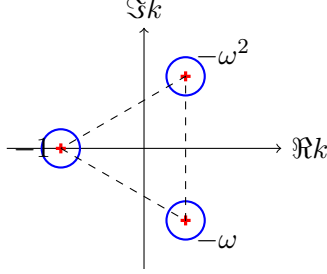


Figure 1: The cyclic cubic family has three exceptional points forming an equilateral triangle. Under $w = k^3 + 1$, each small meridian maps to a positive meridian around 0 in the w -plane.

The eigenvalue equation is

$$E^3 = w, \quad w = k^3 + 1.$$

Therefore the spectral configuration map factors as

$$X \xrightarrow{w=k^3+1} \mathbb{C}^* \xrightarrow{w \mapsto \{w^{1/3}, \omega w^{1/3}, \omega^2 w^{1/3}\}} \text{UConf}_3(\mathbb{C}).$$

This factorisation makes the global braid monodromy explicit.

Proposition 8.1 (Global braid monodromy of the cyclic cubic family). *Let g_0, g_1, g_2 be positive meridians around $-1, -\omega, -\omega^2$, based at a common point and ordered by any certified meridian graph. Under the map $w = k^3 + 1$, each g_j has winding number one around $0 \in \mathbb{C}^*$. Consequently each generator maps to the same local cubic braid*

$$b = \sigma_1 \sigma_2 \in B_3$$

up to the simultaneous conjugacy determined by the base paths. The large positive circle at infinity maps to

$$b^3 = (\sigma_1 \sigma_2)^3,$$

the full twist in B_3 .

Proof. At each zero a_j of $k^3 + 1$, the derivative $3a_j^2$ is nonzero, so a small positive meridian around a_j maps to a positive meridian around 0 in the w -plane. For $E^3 = w$, one positive circuit of w around zero sends the three local branches cyclically. With Convention 2.3 (equivalently, the standard local braid convention), this is represented by $\sigma_1 \sigma_2$. For a large circle $k = R e^{it}$, one has $w = k^3 + 1 = R^3 e^{3it} + 1$, so for large R the image winds three times around zero. Hence the braid at infinity is the third power of the local cubic braid. $\square$

Proposition 8.2 (Image of the cyclic cubic monodromy). *For the cyclic cubic family, the braid-monodromy representation factors as*

$$\pi_1(\mathbb{C} \setminus \{-1, -\omega, -\omega^2\}) \longrightarrow \pi_1(\mathbb{C}^*) \cong \mathbb{Z} \longrightarrow B_3,$$

where the last map sends 1 to $b = \sigma_1 \sigma_2$. Hence

$$\text{Im } \beta_{H_3} = \langle \sigma_1 \sigma_2 \rangle \cong \mathbb{Z}.$$

The induced permutation monodromy has image the cyclic group $\mathbb{Z}/3\mathbb{Z}$.

Proof. The factorisation through $w = k^3 + 1$ was established above. Each finite meridian has winding number one in $\mathbb{C}^*$, and the cubic-root spectral cover sends one positive winding to $b = \sigma_1\sigma_2$. The element $\sigma_1\sigma_2$ has infinite order in B_3 , since its cube is the nontrivial central full twist. Its image in the symmetric group is the three-cycle (123), which has order three. $\square$

Remark 8.3 (Factorisation versus generic monodromy). *This example is global but not maximally noncommutative: all finite meridians have the same image in the cyclic subgroup generated by $\sigma_1\sigma_2$. Generic perturbations of the matrix family may break this factorisation while the graph certificate still applies without modification.*

8.1 Certified radius on a meridian graph

Let $G \subset X$ be any compact meridian graph. The exact discriminant margin is

$$M_G = 27 \inf_{k \in G} |k^3 + 1|^2.$$

If only samples are available, the certified margin is

$$M_G^{\text{cert}} = \widehat{M}_G - \mathcal{G}_S(R_*) L_G h_G.$$

Because $H_3(k)$ is trace-zero, trace-zero perturbations may use the exact constant

$$\mathcal{G}_{\text{tr}=0,3}(R_*) = 162R_*^5.$$

If no trace constraint is imposed on the perturbation, one must instead use an unstructured cubic constant.

For the simple norm bound, note that each row and each column of $H_3(k)$ has absolute row or column sum at most $|k| + 1$. Thus

$$\|H_3(k)\|_1 = \|H_3(k)\|_\infty \leq |k| + 1, \quad \|H_3(k)\|_2 \leq \sqrt{\|H_3(k)\|_1 \|H_3(k)\|_\infty} \leq |k| + 1,$$

by the standard interpolation inequality. Hence one can take

$$R_* > 1 + \sup_{k \in G} |k|.$$

A trace-zero perturbation $\tilde{H}_3 : G \rightarrow \{A : \text{tr } A = 0\}$ is certified to have the same meridian braids whenever

$$\sup_{k \in G} \left\| \tilde{H}_3(k) - H_3(k) \right\|_2 < \frac{M_G^{\text{cert}}}{2 \cdot 162R_*^5}.$$

Example 8.4 (Three equal circular meridians). *Let G be formed by three small circles*

$$k = a_j + re^{it}, \quad 0 \leq t \leq 2\pi,$$

joined to a basepoint by arcs staying outside the disks. On each small circle,

$$k^3 + 1 = 3a_j^2 re^{it} + O(r^2),$$

and hence

$$|\Delta(k)| = 27 |k^3 + 1|^2 = 243r^2 + O(r^3).$$

Thus the certified radius near a small meridian scales like r^2/R_^5 for operator-norm discriminant certificates. The braid is nontrivial for every $r > 0$, but the radius collapses quadratically as the meridian approaches the exceptional point.*

9 Local models and interpretation

The two cubic arrangements above are the main examples. This final section records, in compressed form, how the same graph certificate interfaces with standard local braid-monodromy models.

Remark 9.1 (Cusp of the versal cubic). *For the depressed cubic*

$$P_{u,v}(E) = E^3 + uE + v, \quad A(u, v) = \begin{pmatrix} 0 & 0 & -v \\ 1 & 0 & -u \\ 0 & 1 & 0 \end{pmatrix},$$

the discriminant is $\Delta(u, v) = -4u^3 - 27v^2$. The cusp $4u^3 + 27v^2 = 0$ is parametrised by $u = -3t^2$, $v = 2t^3$. Projecting to the u -line, the two branch values in the v -fibre are

$$v_{\pm}(u) = \pm \frac{2}{3\sqrt{3}}(-u)^{3/2}.$$

When u winds once around zero, the relative vector $v_+(u) - v_-(u)$ rotates through angle 3π , so the discriminant braid is $\tau^3 \in B_2$. This is the classical braid monodromy of the plane cusp [1, 5]. Any compact graph in the complement of the cusp also carries the finite-matrix discriminant-margin certificate of Theorem 4.2. The one-parameter companion cubic $E^3 - 3E - k$ is the slice $u = -3$, $v = -k$, where the cusp has split into the two simple branch values $k = \pm 2$.

Proposition 9.2 (A quartic A_3 arrangement with certified radius). *For $Q_k(E) = E^4 - 4E - k$, the derivative $4(E^3 - 1)$ has critical points $1, \omega, \omega^2$ with branch values $k_j = -3\omega^j$ ($j = 0, 1, 2$), and*

$$\text{Disc}(Q_k) = -256(k^3 + 27).$$

The polynomial $E^4 - 4E$ is the standard A_3 Morsification [1]. In a distinguished A_3 Lefschetz basis the local monodromies at the three branch values are the adjacent half-twists

$$\beta(g_1) = \sigma_1, \quad \beta(g_2) = \sigma_2, \quad \beta(g_3) = \sigma_3 \in B_4,$$

forming the A_3 Dynkin chain $\sigma_1 - \sigma_2 - \sigma_3$. The braid at infinity is $\beta(g_{\infty}) = (\sigma_1 \sigma_2 \sigma_3)^{-1}$.

For the meridian $k = -3 + \rho e^{it}$ around the branch value $k_0 = -3$, the discriminant margin satisfies

$$M_{\rho} = 256 \min_{|k+3|=\rho} |k^3 + 27| = 256 \cdot \min_{|w|=\rho} |w^3 - 9w^2 + 27w| \approx 256 \cdot 27\rho \quad (\rho \rightarrow 0).$$

The companion matrix of Q_k is traceless (its diagonal is zero), so the traceless $n = 4$ discriminant Lipschitz estimate applies. Using the unconditional matrix-gradient bound $G_4^{\text{tr}=0}(R) \leq (60416/3)R^{11}$, the certified perturbation radius is

$$\varepsilon_{\rho}^{\nabla} = \frac{M_{\rho}}{2(60416/3)R_{\rho}^{11}},$$

where $R_{\rho} = \sup_{|k+3| \leq \rho} \|A_k\|_2$. The reduced Pick recursion in Lemma 3.9 and the boundary certificates in Proposition 3.10 improve this to

$$\varepsilon_{\rho}^{\text{sharp}} = \frac{M_{\rho}}{6144R_{\rho}^{11}},$$

a factor $60416/(3 \cdot 3072)$ over the unconditional matrix-gradient radius.

Remark 9.3 (Braid at infinity and van Kampen consistency). *For a punctured plane, compactification to the Riemann sphere adds a meridian g_∞ satisfying*

$$g_1 \cdots g_m g_\infty = 1.$$

The induced braid representation sends this relation to the identity. For the non-abelian cubic one obtains $\beta(g_\infty) = (\sigma_1 \sigma_2)^{-1}$, in agreement with the large-parameter asymptotics $E^3 \sim k$. The graph certificate keeps this braid at infinity and the finite meridian braids fixed under the same operator-norm perturbation bound.

Proposition 9.4 (Simple branch values and certified half-twists). *Let $H(k)$ be analytic near $k = a$. Suppose that $H(a)$ has exactly one repeated eigenvalue of algebraic multiplicity two, with a single Jordan block on that two-dimensional generalized eigenspace, and all other eigenvalues are simple and separated. Then a sufficiently small positive meridian about a has spectral braid equal to a simple half-twist, up to projection and orientation convention. If the meridian has a certified positive discriminant margin, this half-twist is stable under all perturbations inside the corresponding graph radius.*

Proof. Weierstrass preparation and the local two-level normal form reduce the colliding pair to $E^2 = s$ after analytic changes of coordinates; the remaining eigenvalues stay separated. A circuit around $s = 0$ exchanges the two local roots by one half-twist. The stability statement is exactly Theorem 4.2 on the small meridian graph. $\square$

Remark 9.5 (Interpretation). *For non-Hermitian matrix models, the eigenvalue braid is a topological charge attached to an exceptional-point arrangement. The certificate above turns that charge into a finite statement: on the selected graph, every listed meridian braid, product relation, and braid at infinity is unchanged below a computable operator-norm threshold. The construction is close in spirit to Picard–Lefschetz and Zariski–van Kampen braid monodromy, but its input and output are spectral: sampled matrices, braid words, a mesh correction, and a perturbation radius.*

Remark 9.6 (Limitations and further directions). *The companion cubic is surjective onto B_3 but not faithful: its kernel is the normal closure of the Artin relation. A sharper example would be a finite matrix family whose spectral braid monodromy embeds a parameter-domain group into an Artin group. On the quantitative side, the main missing ingredient beyond cubics is a supply of sharp structured discriminant constants. The coefficient-box bound of Proposition 3.22 works in every degree, but it becomes conservative quickly as the resultant degree and the norm scale grow. A hybrid implementation should use the polynomial discriminant certificate near exceptional points and the Bauer–Fike gap certificate on well-conditioned parts of the graph. The two-parameter Brillouin torus extension is developed in Section 10.*

10 Two-parameter braid monodromy on the Brillouin torus

The graph certificate of Section 4 is dimension-free: it uses only a pointwise discriminant Lipschitz estimate and homotopy invariance into configuration space, so it applies to any compact parameter space. This section specialises to the two-dimensional Brillouin torus $\mathbb{T}^2 = S^1 \times S^1$, which is the natural parameter space for two-band non-Hermitian lattice models. The main content is a precise identification of the discriminant locus, two versions of the stability theorem, and an explicit example.

10.1 Discriminant locus on $\mathbb{T}^2$

Write $z_j = e^{ik_j}$ and let $H(k_1, k_2)$ be a finite-Fourier analytic matrix family on $\mathbb{T}^2$. The discriminant

$$\Delta(k_1, k_2) = \text{Disc}(H(k_1, k_2))$$

is a complex-valued real-analytic function on $\mathbb{T}^2$. Two cases arise.

Generic complex case. For a general non-Hermitian family, $\Delta : \mathbb{T}^2 \rightarrow \mathbb{C}$ is complex-valued. The vanishing set $C = \{(k_1, k_2) : \Delta(k_1, k_2) = 0\}$ imposes two real equations on the real two-dimensional manifold $\mathbb{T}^2$, so generically C is a *finite set of isolated points*. On the complexified parameter torus $(\mathbb{C}^*)^2$, the zero locus $\mathcal{C}_{\mathbb{C}}$ is a complex algebraic curve; its intersection with the real torus $|z_1| = |z_2| = 1$ is generically finite.

Real-discriminant case. If Δ is real-valued on $\mathbb{T}^2$ —for example, when the matrix family has a reality symmetry making eigenvalues come in conjugate pairs, or when a scalar factor forces the discriminant to be real—then $\Delta = 0$ is a single real equation and $C \subset \mathbb{T}^2$ is generically an embedded real-analytic curve. Each connected component of C is a circle with a primitive homology class $(p, q) \in H_1(\mathbb{T}^2; \mathbb{Z}) \cong \mathbb{Z}^2$, $\gcd(p, q) = 1$, or a null-homologous oval.

For a traceless $n = 3$ family with $\Delta = -4p^3 - 27q^2$, where $p = -\frac{1}{2} \text{tr}(H^2)$ and $q = -\det H$ are Laurent polynomials of bidegree $\leq (2d_1, 2d_2)$ and $(3d_1, 3d_2)$ respectively, the discriminant has bidegree at most $(6d_1, 6d_2)$. In the real-discriminant setting this bounds the topological complexity of C via Harnack-type estimates.

10.2 Fundamental groups

The relevant fundamental group depends on the type of locus.

Finite-point locus (m isolated points).

$$\pi_1(\mathbb{T}^2 \setminus C) \cong \langle a, b, \mu_1, \dots, \mu_m \mid [a, b] \mu_1 \cdots \mu_m = 1 \rangle,$$

where a, b are the two standard torus generators and each μ_j is a small meridian around the j -th exceptional point. After eliminating one generator this is the free group F_{m+1} . The two ambient torus cycles a, b appear alongside the exceptional-point meridians, carrying additional braid monodromy absent from the one-parameter punctured-plane setting.

Embedded curve locus (one component (p, q) -curve). If C is a single embedded essential (p, q) -curve, then $\mathbb{T}^2 \setminus C$ is an open annulus and $\pi_1(\mathbb{T}^2 \setminus C) \cong \mathbb{Z}$.

Zariski-van Kampen presentation. In general, project $\pi : \mathbb{T}^2 \rightarrow S^1$ by $\pi(k_1, k_2) = k_1$. Over a regular value the fiber is S^1 punctured at d points of $C \cap \pi^{-1}(k_1)$, giving fiber group $\pi_1 \cong F_d$. The monodromy of these punctures as k_1 traverses the finitely many critical values of $\pi|_C$ supplies the remaining relations:

$$\pi_1(\mathbb{T}^2 \setminus C) = \langle \pi_1(F_{k_1^0}), \gamma_1, \dots, \gamma_\ell, \eta \mid \gamma_j x \gamma_j^{-1} = \rho_j(x), \eta x \eta^{-1} = \rho_\infty(x), \gamma_1 \cdots \gamma_\ell = \eta \rangle,$$

where ρ_j is the fiber braid monodromy at the j -th critical value and η records the total monodromy of the base S^1 .

10.3 Stability theorems

Two distinct stability statements arise, according to whether $M_{\mathbb{T}^2} > 0$.

Theorem 10.1 (Full-torus stability). *Let $H : \mathbb{T}^2 \rightarrow \mathcal{S} \subset M_n(\mathbb{C})$ be continuous, with $\sup_{\mathbb{T}^2} \|H(k)\|_2 \leq R$ and*

$$M_{\mathbb{T}^2} := \inf_{k \in \mathbb{T}^2} |\Delta(k)| > 0.$$

Choose $R_* > R$ and let $\mathcal{G}_S(R_*)$ be an admissible discriminant Lipschitz constant on $\mathcal{S} \cap \{\|A\|_2 \leq R_*\}$. If

$$\sup_{\mathbb{T}^2} \|\tilde{H}(k) - H(k)\|_2 < \min \left\{ R_* - R, \frac{M_{\mathbb{T}^2}}{2\mathcal{G}_S(R_*)} \right\},$$

then $H_t = (1-t)H + t\tilde{H}$ has simple spectrum on all of $\mathbb{T}^2$ for every $t \in [0, 1]$, and the spectral braid representations $\phi_H, \phi_{\tilde{H}} : \pi_1(\mathbb{T}^2) \rightarrow B_n$ are equal up to simultaneous conjugacy.

Proof. The Lipschitz estimate gives, pointwise in k , $|\Delta(H_t(k)) - \Delta(H(k))| \leq \mathcal{G}_S(R_*) \sup \|\tilde{H} - H\|_2 < M_{\mathbb{T}^2}/2$, so $|\Delta(H_t(k))| \geq M_{\mathbb{T}^2}/2 > 0$ for all t, k . The map $(t, k) \mapsto \Lambda_{H_t}(k)$ is therefore a homotopy of $\mathbb{T}^2 \rightarrow \text{UConf}_n(\mathbb{C})$. Homotopic maps induce equal representations on π_1 , up to the basepoint-identification conjugacy. $\square$

Remark 10.2. $M_{\mathbb{T}^2} > 0$ is equivalent to $C = \emptyset$. When $C \neq \emptyset$, $M_{\mathbb{T}^2} = 0$ and Theorem 10.1 does not apply. The correct object is then a positive margin on a compact $K \subset \mathbb{T}^2 \setminus C$.

Theorem 10.3 (Torus graph stability). *Let $K \subset \mathbb{T}^2 \setminus C$ be a compact subset whose inclusion induces a surjection $\pi_1(K) \twoheadrightarrow \pi_1(\mathbb{T}^2 \setminus C)$, for example a deformation retract or a carrying graph. Assume $\sup_K \|H(k)\|_2 \leq R$ and $M_K := \inf_K |\Delta(k)| > 0$. If*

$$\sup_K \|\tilde{H}(k) - H(k)\|_2 < \min \left\{ R_* - R, \frac{M_K}{2\mathcal{G}_S(R_*)} \right\},$$

then the induced representations $\phi_H, \phi_{\tilde{H}} : \pi_1(K) \rightarrow B_n$ are equal up to simultaneous conjugacy; and if $K \simeq \mathbb{T}^2 \setminus C$, then so are the full complement representations.

Proof. This is exactly Theorem 4.2 with $\mathcal{P} = \mathbb{T}^2$. The proof is pointwise in matrix space and does not depend on the dimension of the parameter space. $\square$

For traceless 3×3 families, both theorems use $\mathcal{G}_{\text{tr}=0,3}(R_*) = 162R_*^5$ from Section 3.

10.4 Equivalence relation for two-parameter braid monodromy

For a one-parameter loop, the invariant is a conjugacy class in B_n . For the two-parameter representation $\phi_H : \pi_1(\mathbb{T}^2 \setminus C, p_0) \rightarrow B_n$, the correct equivalence is *simultaneous conjugacy*: two representations $\phi, \tilde{\phi}$ are equivalent if there exists $b \in B_n$ with

$$\tilde{\phi}(\gamma) = b \phi(\gamma) b^{-1} \quad \text{for all } \gamma \in \pi_1(\mathbb{T}^2 \setminus C, p_0).$$

If the basepoint or generating system changes, one additionally precomposes with the induced automorphism of $\pi_1(\mathbb{T}^2 \setminus C)$. This is strictly stronger than checking individual braid words: the representation records non-abelian relations between the two ambient torus cycles, the exceptional-point meridians, and any torus relation they satisfy.

10.5 Two-parameter Hatano–Nelson example

Let $M = J_\omega = \text{diag}(1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, and define

$$H(k_1, k_2) = f(k_1, k_2)I + g(k_1, k_2)M,$$

where f, g are Laurent polynomials. Since eigenvalue differences scale by g , we have

$$\Delta(k_1, k_2) = \text{Disc}(M) \cdot g(k_1, k_2)^6 = -27 g(k_1, k_2)^6.$$

Example A (real Δ , embedded curve C). Take

$$g(k_1, k_2) = m + t_1 \cos k_1 + t_2 \cos k_2, \quad m, t_1, t_2 \in \mathbb{R}.$$

Then $\Delta \in \mathbb{R}$ on $\mathbb{T}^2$ and $C = \{m + t_1 \cos k_1 + t_2 \cos k_2 = 0\}$. If $|m| > |t_1| + |t_2|$ then $C = \emptyset$ and $M_{\mathbb{T}^2} = 27 \inf |g|^6 > 0$; Theorem 10.1 gives an explicit radius with $\mathcal{G}_{\text{tr}=0,3}(R_*) = 162R_*^5$. If $|m| < |t_1| + |t_2|$ then C is a non-empty embedded curve; $\pi_1(\mathbb{T}^2 \setminus C)$ is a free group, and the graph certificate of Theorem 10.3 applies on any compact carrying subgraph.

Example B (non-real Δ , finite C). Take

$$g(k_1, k_2) = ae^{ik_1} + be^{-ik_1} + ce^{ik_2} - \gamma, \quad a \neq b.$$

Then $\Delta = -27g^6$ is complex-valued. Its zero set $C = \{g = 0\} \subset \mathbb{T}^2$ is generically a pair of isolated exceptional points. The group $\pi_1(\mathbb{T}^2 \setminus C) \cong F_3$ has generators a, b, μ_1, μ_2 with relation $[a, b]\mu_1\mu_2 = 1$. Since all eigenvalues of $H = gM$ rotate rigidly together, every generator maps into the centre $Z(B_3)$; the image is a cyclic group. Explicit computations appear in Propositions 10.4–10.5 below.

Proposition 10.4 (Full-torus monodromy, no exceptional points). *Let $M = \text{diag}(1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, $H(k_1, k_2) = g(k_1, k_2)M$ with*

$$g(k_1, k_2) = e^{ik_1} + 0.3e^{-ik_1} + 0.5e^{ik_2}.$$

Since $|e^{ik_1} + 0.3e^{-ik_1}|^2 = 1.09 + 0.6 \cos(2k_1) \geq 0.49$, we have $|e^{ik_1} + 0.3e^{-ik_1}| \geq 0.7 > 0.5$ for all k_1 , so g has no zeros on $\mathbb{T}^2$ and $M_{\mathbb{T}^2} > 0$.

The spectral braid monodromy $\phi : \pi_1(\mathbb{T}^2) \cong \mathbb{Z}^2 \rightarrow B_3$ is computed from the winding numbers of g along the two standard cycles. Along $k_1 : 0 \rightarrow 2\pi$ with $k_2 = 0$ fixed, $g = e^{ik_1} + 0.3e^{-ik_1} + 0.5$ has winding number $w_1 = 1$ around 0. Along $k_2 : 0 \rightarrow 2\pi$ with $k_1 = 0$, $g = 1.3 + 0.5e^{ik_2}$ satisfies $|g| \geq 0.8 > 0$, so $w_2 = 0$. A winding-number- w loop of g rotates all three eigenvalues $g, g\omega, g\omega^2$ rigidly by $2\pi w$, corresponding to the w -th power of the positive full twist $\Omega = (\sigma_1\sigma_2)^3 \in Z(B_3)$. Therefore

$$\phi(\alpha_1) = (\sigma_1\sigma_2)^3, \quad \phi(\alpha_2) = 1, \quad \phi(m, n) = ((\sigma_1\sigma_2)^3)^m.$$

The image is $\langle (\sigma_1\sigma_2)^3 \rangle \cong \mathbb{Z}$, a cyclic subgroup of $Z(B_3)$. Theorem 10.1 applies with $\mathcal{G}_{\text{tr}=0,3}(R_) = 162R_*^5$.*

Proposition 10.5 (Punctured-torus monodromy, two exceptional points). *Keep $H = gM$ with*

$$g(k_1, k_2) = e^{ik_1} + 0.3e^{-ik_1} + 0.5e^{ik_2} - 0.8.$$

The equation $g = 0$ on $\mathbb{T}^2$ has exactly two transverse zeros, located at

$$\begin{aligned} p_1 &= \left(\arccos \frac{11}{15}, \pi + \arccos \frac{23}{75} \right) \approx (0.7476, 4.4007), \\ p_2 &= \left(2\pi - \arccos \frac{11}{15}, \pi - \arccos \frac{23}{75} \right) \approx (5.5356, 1.8825), \end{aligned}$$

with Jacobians

$$J(p_1) = -\frac{8\sqrt{26}}{375} < 0, \quad J(p_2) = +\frac{8\sqrt{26}}{375} > 0.$$

The fundamental group $\pi_1(\mathbb{T}^2 \setminus \{p_1, p_2\}) \cong F_3$ has presentation $\langle a, b, \mu_1, \mu_2 \mid [a, b]\mu_1\mu_2 = 1 \rangle$. Setting $\Omega = (\sigma_1\sigma_2)^3$, the braid monodromy representation $\beta : \pi_1(\mathbb{T}^2 \setminus \{p_1, p_2\}) \rightarrow B_3$ is

$$\beta(a) = \Omega, \quad \beta(b) = 1, \quad \beta(\mu_1) = \Omega^{-1}, \quad \beta(\mu_2) = \Omega.$$

The image is $\langle \Omega \rangle \cong \mathbb{Z} \subset Z(B_3)$.

Proof. Write $c = \cos k_1$, $s = \sin k_1$, $C = \cos k_2$, $S = \sin k_2$. The condition $g = 0$ is equivalent to the real system

$$1.3c + 0.5C = 0.8, \quad 0.7s + 0.5S = 0.$$

The second equation gives $S = -1.4s$; substituting $C = (0.8 - 1.3c)/0.5$ into $C^2 + S^2 = 1$ and expanding with $s^2 = 1 - c^2$ yields

$$15c^2 - 26c + 11 = (15c - 11)(c - 1) = 0,$$

so $c \in \{1, 11/15\}$.

Case $c = 1$ ($k_1 = 0$, $S = 0$, $C = -1$, $k_2 = \pi$). The Jacobian of the real map $(k_1, k_2) \mapsto (\operatorname{Re} g, \operatorname{Im} g)$ is

$$J(k_1, k_2) = -0.65 \sin k_1 \cos k_2 + 0.35 \sin k_2 \cos k_1.$$

At $(0, \pi)$: $J = 0$. A local expansion gives $g(\epsilon, \pi + \delta) \approx (-0.65\epsilon^2 + 0.25\delta^2) + i(0.7\epsilon + 0.5\delta)$, so on the circle $(\epsilon, \delta) = r(\cos \theta, \sin \theta)$ the image of g stays near the imaginary axis to leading order and does not wind around the origin; the local degree of g at $(0, \pi)$ is zero. This zero therefore carries no braid monodromy and is excluded from the exceptional-point count.

Case $c = 11/15$. Then $s = \pm 2\sqrt{26}/15$, $C = -23/75$, $S = \mp 14\sqrt{26}/75$. (One checks $C^2 + S^2 = (529 + 196 \cdot 26)/75^2 = 1$.) The two solutions are p_1 (sign $+$) and p_2 (sign $-$), with coordinates as stated. The Jacobian formula gives

$$\begin{aligned} J(p_{1,2}) &= -0.65 \cdot \left(\pm \frac{2\sqrt{26}}{15}\right) \cdot \left(-\frac{23}{75}\right) + 0.35 \cdot \left(\mp \frac{14\sqrt{26}}{75}\right) \cdot \frac{11}{15} \\ &= \frac{\sqrt{26}}{22500} (\pm 598 \mp 1078) = \mp \frac{8\sqrt{26}}{375}, \end{aligned}$$

confirming $\operatorname{sgn} J(p_1) = -1$ and $\operatorname{sgn} J(p_2) = +1$.

Braid monodromy. Since $H = gM$ and the eigenvalues of $M = \operatorname{diag}(1, \omega, \omega^2)$ are distinct and fixed, all three eigenvalues of H rotate rigidly by g . By the same winding-number argument as in Proposition 10.4, the torus generators give $\beta(a) = \Omega$ and $\beta(b) = 1$. For the meridian μ_j around p_j , the local degree of g at p_j equals $\operatorname{sgn} J(p_j)$; a winding-number- w loop of g produces braid $\Omega^w \in Z(B_3)$. Hence $\beta(\mu_1) = \Omega^{-1}$ and $\beta(\mu_2) = \Omega$.

The Zariski–van Kampen relation $\beta([a, b])\beta(\mu_1)\beta(\mu_2) = 1$ reduces to $[\Omega, 1] \cdot \Omega^{-1} \cdot \Omega = 1$, which holds since $\Omega \in Z(B_3)$. This is consistent: the abelian image forces all ZvK relations to be automatic, reflecting that the representation factors through $\pi_1^{\text{ab}} \cong H_1(\mathbb{T}^2 \setminus \{p_1, p_2\})$.

Contrast with the companion cubic (Section 7): for $E^3 - 3E - k$, the meridians map to $\sigma_1, \sigma_2 \notin Z(B_3)$ and the image is all of B_3 . Here every generator maps into the centre because the $H = gM$ structure propagates the winding of the scalar factor g to all strands simultaneously. $\square$

A Takenaka–Malmquist entries for the quartic gradient

Let $\ell_i = \lambda_i$ and $d_i = (1 - |\lambda_i|^2)^{1/2}$. For the rank-one defect model S_B , the upper triangular entries of

$$A_B = c_1 S_B + c_2 S_B^2 + c_3 S_B^3$$

are

$$\begin{aligned}
(A_B)_{11} &= \ell_1(c_1 + c_2\ell_1 + c_3\ell_1^2), \\
(A_B)_{12} &= d_1d_2\{c_1 + c_2(\ell_1 + \ell_2) + c_3(\ell_1^2 + \ell_1\ell_2 + \ell_2^2)\}, \\
(A_B)_{13} &= -d_1d_3\{c_1\ell_2 - c_2d_2^2 + c_2\ell_1\ell_2 + c_2\ell_2\ell_3 \\
&\quad - c_3d_2^2(\ell_1 + \ell_2 + \ell_3) + c_3\ell_1^2\ell_2 + c_3\ell_1\ell_2\ell_3 + c_3\ell_2\ell_3^2\}, \\
(A_B)_{14} &= d_1d_4\{c_1\ell_2\ell_3 - c_2d_2^2\ell_3 - c_2d_3^2\ell_2 + c_2\ell_1\ell_2\ell_3 + c_2\ell_2\ell_3\ell_4 \\
&\quad + c_3d_2^2d_3^2 - c_3d_2^2(\ell_1\ell_3 + \ell_2\ell_3 + \ell_3\ell_4) \\
&\quad - c_3d_3^2(\ell_1\ell_2 + \ell_2\ell_3 + \ell_2\ell_4) + c_3\ell_1^2\ell_2\ell_3 \\
&\quad + c_3\ell_1\ell_2\ell_3\ell_4 + c_3\ell_2\ell_3\ell_4^2\}, \\
(A_B)_{22} &= \ell_2(c_1 + c_2\ell_2 + c_3\ell_2^2), \\
(A_B)_{23} &= d_2d_3\{c_1 + c_2(\ell_2 + \ell_3) + c_3(\ell_2^2 + \ell_2\ell_3 + \ell_3^2)\}, \\
(A_B)_{24} &= -d_2d_4\{c_1\ell_3 - c_2d_3^2 + c_2\ell_2\ell_3 + c_2\ell_3\ell_4 \\
&\quad - c_3d_3^2(\ell_2 + \ell_3 + \ell_4) + c_3\ell_2^2\ell_3 + c_3\ell_2\ell_3\ell_4 + c_3\ell_3\ell_4^2\}, \\
(A_B)_{33} &= \ell_3(c_1 + c_2\ell_3 + c_3\ell_3^2), \\
(A_B)_{34} &= d_3d_4\{c_1 + c_2(\ell_3 + \ell_4) + c_3(\ell_3^2 + \ell_3\ell_4 + \ell_4^2)\}, \\
(A_B)_{44} &= \ell_4(c_1 + c_2\ell_4 + c_3\ell_4^2).
\end{aligned}$$

B Pick certificate data for the quartic Blaschke model

This appendix gives the finite calculation associated with Remark 3.8. The centred constraint is imposed by

$$\lambda_4 = -(\lambda_1 + \lambda_2 + \lambda_3).$$

Let $q_i = q_\lambda(\lambda_i)$, where

$$q_\lambda(z) = 768^{-1}(c_1z + c_2z^2 + c_3z^3).$$

The four values q_i are homogeneous polynomials of degree 11 divided by 192. For example,

$$q_1 = \frac{\lambda_1}{192} Q_1(\lambda_1, \lambda_2, \lambda_3),$$

where Q_1 is a degree-ten integer polynomial. The other q_i are obtained by permuting the centred roots.

The reduced data used in the non-normal certificate are

$$r_i = \frac{q_i}{\lambda_i} = 768^{-1}(c_1 + c_2\lambda_i + c_3\lambda_i^2),$$

with the limiting value understood when $\lambda_i = 0$. The reduced Pick matrix is

$$\Pi_\lambda^{\text{red}} = \left[\frac{1 - r_i \overline{r_j}}{1 - \lambda_i \overline{\lambda_j}} \right]_{i,j=1}^4.$$

The verification is carried out on principal submatrices of Π_λ^{red} . For a fixed ordering α and a principal index set I , use the Schur complement recursion

$$\Pi^{(1)} = \Pi, \quad \Pi^{(m+1)} = \Pi_{22}^{(m)} - \Pi_{21}^{(m)} (\Pi_{11}^{(m)})^{-1} \Pi_{12}^{(m)},$$

with the rows and columns ordered by α . Clearing the positive denominators

$$D_m = \prod_{i,j \leq m} (1 - \lambda_i \overline{\lambda_j}) \prod_{p < q} |\lambda_p - \lambda_q|^{2e_m}$$

gives scalar real polynomials $R_{I,\alpha,m}$ after the substitutions

$$\lambda_j = x_j + iy_j, \quad \overline{\lambda_j} = x_j - iy_j, \quad x_4 + iy_4 = -(\lambda_1 + \lambda_2 + \lambda_3).$$

Thus

$$R_{I,\alpha,m} \in \mathbb{Z}[x_1, y_1, x_2, y_2, x_3, y_3].$$

The sign condition for the reduced Pick matrix is exactly

$$R_{I,\alpha,m} \geq 0$$

on the semialgebraic chamber

$$x_j^2 + y_j^2 \leq 1, \quad (x_i - x_j)^2 + (y_i - y_j)^2 > 0, \quad |\lambda_{\alpha(1)}| \geq \cdots \geq |\lambda_{\alpha(4)}|.$$

For the compact Bernstein check, rotate so that $\lambda_{\alpha(1)} = \rho$ with $0 \leq \rho \leq 1$, write the remaining two free roots as

$$\lambda_{\alpha(2)} = \rho_2 e^{i\theta_2}, \quad \lambda_{\alpha(3)} = \rho_3 e^{i\theta_3},$$

and set

$$X_0 = \rho, \quad X_1 = \rho_2^2, \quad X_2 = \rho_3^2, \quad X_3 = \cos \theta_2, \quad X_4 = \sin \theta_2, \quad X_5 = \cos \theta_3, \quad X_6 = \sin \theta_3.$$

The two circle relations $X_3^2 + X_4^2 = 1$, $X_5^2 + X_6^2 = 1$ are eliminated boxwise by the tangent half-angle substitution on four angular charts. Each chamber polynomial therefore becomes a finite list of ordinary real polynomials on boxes $[0, 1]^6$. Bernstein subdivision is valid because every denominator removed above is strictly positive on the interior of the chamber, and the boundary boxes are precisely the normal, 3+1, paired, radial-square, or confluent faces already certified in the main text.

The exact-arithmetic certificate is therefore a finite table: for each tuple (I, α, m, ν) , where ν indexes the chamber and angular chart, list the Bernstein degree vector $d_{I,\alpha,m,\nu}$ and the minimum Bernstein coefficient

$$\beta_{I,\alpha,m,\nu}^{\min} \geq 0.$$

Strictly positive $\beta^{\min}$ proves the corresponding box immediately; zero coefficients are admissible only on one of the already certified boundary faces. This is the exact form in which the remaining interior certificate is finite and machine-checkable.

Remark B.1 (Why the resolved certificate is used). *The direct five-real-variable Bernstein table is finite but too large to serve as a useful printed certificate. Expanding only the 2×2 reduced Pick minor numerators in the centred variables*

$$\lambda_1 = r_1, \quad \lambda_2 = x_2 + iy_2, \quad \lambda_3 = x_3 + iy_3, \quad \lambda_4 = -(\lambda_1 + \lambda_2 + \lambda_3)$$

already gives total degree 42. The six unordered pairs have respectively

$\{i, j\}$	12	13	14	23	24	34
# monomials	100400	100400	101780	103234	103210	103210.

Thus the brute-force table is a legitimate finite verification problem, but not a practical appendix-level proof. The proof above uses the second option: the lower-dimensional Schur-pivot certificate isolates the radial-square degeneration, and Appendix E resolves the only hard Bernstein corner by a blow-up.

Remark B.2 (Why this is not an infinite regress). *The reduction stops at this point. No new analytic or functional-analytic statement remains after Proposition 3.19: the objects to be checked are explicitly generated polynomials with rational coefficients on rational boxes. The required certificates are supplied here: the regular boxes are covered by the Schur-pivot Bernstein tables, and the unique hard corner is covered by the blow-up certificate in Appendix E. Therefore the non-normal quartic case is closed, rather than merely reduced to a further condition.*

Lemma B.3 (Bernstein certificate implication). *Let $P \in \mathbb{Q}[t_1, \dots, t_s]$ and let $Q = \prod_a q_a$ be a product of constraint polynomials which are non-negative on a box $B \subset [0, 1]^s$. Suppose that, after subdivision of B into finitely many rational boxes B_ν , one has an identity*

$$P = B_\nu^{(0)} + \sum_a q_a M_{a,\nu}$$

on B_ν ,

where $B_\nu^{(0)}$ and every multiplier $M_{a,\nu}$ have non-negative Bernstein coefficients on B_ν . Then $P \geq 0$ on B . If the only zero Bernstein coefficients occur on faces contained in the union of the certified boundary strata, then $P > 0$ on the genuinely interior non-confluent part of B .

Proof. The Bernstein basis is a partition of unity by non-negative polynomials on a box. Therefore a polynomial with non-negative Bernstein coefficients is non-negative throughout that box. Since $q_a \geq 0$ and $M_{a,\nu} \geq 0$, the displayed identity gives $P \geq 0$ on B_ν . The sign of P on the whole chamber follows after summing over the finite subdivision. The strict statement follows because a Bernstein zero can persist in the interior only if every active basis term carrying that point has zero coefficient; by the certificate table these zero terms are supported only on the already certified boundary faces. $\square$

C Pair-defect Blaschke certificate

The reduced Pick problem is closed on the whole paired locus. After a common rotation, write a paired centred spectrum as

$$\{1, -1, z, -z\}, \quad |z| \leq 1.$$

For the reduced data

$$r_i = 768^{-1}(c_1 + c_2 \lambda_i + c_3 \lambda_i^2)$$

one obtains

$$\begin{aligned} r(1) = r(-1) = A(z) &:= \frac{z^2(1-z^2)^3(5-z^2)}{48}, \\ r(z) = r(-z) = B(z) &:= \frac{(1-z^2)^3(1-5z^2)}{48}. \end{aligned}$$

The Schur algorithm for the data $(1, A), (-1, A), (z, B), (-z, B)$ gives

$$A, \quad 0, \quad -\frac{B-A}{1-\overline{A}B}, \quad 0.$$

Thus the reduced Pick matrix is positive semidefinite as soon as $|A|, |B| \leq 1$. For $|z| \leq 1$,

$$|A| \leq \frac{|z|^2|1-z^2|^3|5-z^2|}{48} \leq \frac{1 \cdot 2^3 \cdot 6}{48} = 1,$$

and

$$|B| \leq \frac{|1 - z^2|^3 |1 - 5z^2|}{48} \leq \frac{2^3 \cdot 6}{48} = 1.$$

Equality in both estimates can occur only at $z^2 = -1$, namely the square $\{1, -1, i, -i\}$. Hence the paired-locus reduced Pick certificate is unconditional.

The adjacent one-defect boundary edge also has a terminating certificate. On the centred 3+1 edge

$$\{1, e^{i\theta}, e^{-i\theta}, -r\}, \quad \cos \theta = \frac{r-1}{2}, \quad 0 \leq r \leq 1,$$

the reduced Schur gaps factor as rational functions of r . The numerators are products of powers of $1 - r$ and the following real factors:

$$\begin{aligned} p_9 &= 4r^9 - 16r^8 - 15r^7 + 81r^6 + 25r^5 - 91r^4 + 71r^3 + 15r^2 - 165r - 165, \\ p_{10} &= 4r^{10} - 20r^9 + r^8 + 96r^7 - 56r^6 - 116r^5 + 162r^4 - 56r^3 - 180r^2 - 219, \\ p_{18} &= 8r^{18} - 36r^{17} - 126r^{16} + 703r^{15} + 160r^{14} - 3193r^{13} \\ &\quad + 2299r^{12} + 945r^{11} - 9684r^{10} + 14733r^9 + 14163r^8 - 6135r^7 \\ &\quad + 24992r^6 + 21897r^5 + 1737r^4 + 24039r^3 + 37836r^2 + 36135r + 36135. \end{aligned}$$

Sturm's theorem gives no roots in $[0, 1]$ for these three polynomials; their endpoint signs are

$$p_9(0), p_9(1) < 0, \quad p_{10}(0), p_{10}(1) < 0, \quad p_{18}(0), p_{18}(1) > 0.$$

The denominator factors have the same Sturm verification and positive endpoint signs. Therefore the reduced Pick matrix is positive on this one-defect edge, with degeneration only at $r = 1$, where it lands on the square.

The first reduced pivot on the full two-angle 3+1 stratum has a finite certificate. Write

$$P(z) = (z + r)(z^3 - rz^2 + urz - u), \quad u = e^{i\chi}, \quad 0 \leq r \leq 1.$$

The interior node is $-r$. Direct substitution in the reduced value

$$r(\lambda) = 768^{-1}(c_1 + c_2\lambda + c_3\lambda^2)$$

gives

$$1 - |r(-r)|^2 = \frac{M(r, \cos \chi)}{36864},$$

where M is a polynomial of bidegree $(18, 4)$ after the change $\cos \chi = 2y - 1$. In the Bernstein basis on $(r, y) \in [0, 1]^2$, all 95 coefficients of $M(r, 2y - 1)$ are non-negative, and the only common zero of the certificate is the square endpoint $(r, y) = (1, 1)$.

It remains to control the three unit boundary nodes. By rotation it suffices to fix one of them at 1. Let the other two unit nodes be $x, y \in \partial\mathbb{D}$ and put

$$s = x + y, \quad xy = s/\bar{s}, \quad |1 + s| \leq 1.$$

The last inequality is exactly the condition that the fourth root $-(1 + x + y)$ lies in $\overline{\mathbb{D}}$. For the selected unit node 1, the same substitution gives

$$1 - |r(1)|^2 = \frac{N(a, b)}{36864(a^2 + b^2)^4}, \quad s = a + ib.$$

Set $s = -1 + \rho e^{i\psi}$ and $\cos \psi = 2y - 1$. Since the denominator is positive away from the square limit $s = 0$, it suffices to prove

$$\mathcal{N}(\rho, y) := N(-1 + \rho(2y - 1), \rho\sqrt{1 - (2y - 1)^2}) \geq 0$$

on $[0, 1]^2$. The square root is harmless because $N(a, b)$ is even in b , so $\mathcal{N}$ is an ordinary polynomial. Its bidegree is $(28, 13)$ and it has 223 non-zero monomial terms. Exact rational Bernstein expansion on the following dyadic subdivision gives:

ρ -interval	y -interval	min Bernstein coefficient
$[0, 1/2]$	$[0, 1]$	$32343111/262144$
$[1/2, 1]$	$[0, 1/2]$	0
$[1/2, 1]$	$[1/2, 1]$	0.

Hence $\mathcal{N} \geq 0$ on the whole square. The zero coefficients in the last two boxes occur only on the endpoint faces corresponding to the square configuration, so the inequality is strict away from that endpoint. Therefore $|r_i| \leq 1$ for all four nodes on the full 3+1 stratum. This proves the first reduced Schur pivot on that stratum. Together with the radial boundary blow-up lemma (Lemma 3.13), the strict unit-node inequalities away from the square imply the full reduced Pick certificate on the general two-angle 3+1 boundary.

There is also a closed two-boundary/two-interior radial edge. For

$$\{1, -1, ir, -ir\}, \quad 0 \leq r \leq 1,$$

the reduced Schur gaps are

$$G_1 = \frac{(1 - r^2)(r^2 + 3)(r^6 + 6r^4 + 9r^2 + 16)(r^{10} + 8r^8 + 18r^6 + 16r^4 + 5r^2 + 48)}{2304},$$

$$G_2 = 1, \quad G_4 = 1,$$

and

$$G_3 = \frac{(1 - r^2)^2(r^2 + 3)(r^6 + 6r^4 + 9r^2 + 16)(5r^6 + 21r^4 + 39r^2 + 47)}{(5r^{18} + 56r^{16} + 236r^{14} + 520r^{12} + 670r^{10} + 520r^8 + 236r^6 + 56r^4 + 5r^2 + 2304)^2}$$

multiplied by the positive factors

$$(5r^8 + 16r^6 + 18r^4 + 8r^2 + 49)(r^{10} + 8r^8 + 18r^6 + 16r^4 + 5r^2 + 48).$$

Thus this radial two-pair edge is positive for $0 \leq r < 1$ and becomes sharp only at the square.

For the extremal two-defect Blaschke submodel with ordered spectrum $\{1, ir, -ir, -1\}$, $0 \leq r \leq 1$, the Takenaka–Malmquist matrix gives

$$\sum_{i \leq j} |(A_B)_{ij}| = -16(1 + r)(1 + r^2)^3(3r^3 - 13r^2 - r - 1).$$

Therefore

$$3072 - \sum_{i \leq j} |(A_B)_{ij}| = 16(r - 1)h_9(r),$$

where

$$h_9(r) = 3r^9 - 7r^8 - 12r^7 - 44r^6 - 78r^5 - 114r^4 - 156r^3 - 172r^2 - 189r - 191.$$

Since all coefficients of $h'_9(r)$ except the first are negative and $h_9(1) = -960 < 0$, Sturm's theorem (or direct monotonicity after one derivative split) gives $h_9(r) < 0$ on $[0, 1]$. Hence the pair-defect submodel satisfies

$$\sum_{i \leq j} |(A_B)_{ij}| \leq 3072,$$

with equality only at $r = 1$.

For the Frobenius check on the pair-defect edge, the same substitution gives

$$\|A_B\|_{\mathbb{F}}^2 = 256(1 + r^2)^6(27r^8 + 30r^6 + 76r^4 + 10r^2 + 1).$$

Consequently

$$1536^2 - \|A_B\|_{\mathbb{F}}^2 = 256(1 - r^2)k_{18}(r),$$

where

$$\begin{aligned} k_{18}(r) = & 27r^{18} + 219r^{16} + 880r^{14} + 2336r^{12} + 4542r^{10} \\ & + 6830r^8 + 8392r^6 + 9048r^4 + 9199r^2 + 9215. \end{aligned}$$

Thus the Frobenius certificate is strict on this submodel for $0 \leq r < 1$ and becomes sharp only at $r = 1$.

D Bernstein–Sturm data for the quartic $(3+1)$ monotonicity

This appendix lists the Bernstein coefficients in $w = (1 + c)/2$ used in the proof that $\partial_c \mathcal{B} \geq 0$.

R

Degree in w : 2.

$$\begin{aligned} b_0(r) &= (r - 1)(730r^9 - 910r^8 - 1355r^7 - 1963r^6 - 63r^5 - 711r^4 + 823r^3 - 377r^2 - 135r - 135), \quad b_0(0) = -135, \\ b_1(r) &= 730r^{10} - 1725r^8 + 620r^6 + 1534r^4 + 242r^2 + 135, \quad b_1(0) = 135, \quad b_1(1) = 1536, \quad \text{no zero in } (0, 1) \\ b_2(r) &= (r + 1)(730r^9 + 910r^8 - 1355r^7 + 1963r^6 - 63r^5 + 711r^4 + 823r^3 + 377r^2 - 135r + 135), \quad b_2(0) = 135, \end{aligned}$$

$-E$

Degree in w : 2.

$$\begin{aligned} b_0(r) &= -(r - 1)(r + 3)^2(2r^2 + r + 1)(2r^5 + 9r^4 - 24r^3 + 17r^2 - 24r + 36), \quad b_0(0) = 324, \quad b_0(1) = 0, \\ b_1(r) &= -4r^{10} - 235r^8 - 69r^6 - r^4 + 369r^2 + 324, \quad b_1(0) = 324, \quad b_1(1) = 384, \quad \text{no zero in } (0, 1) \\ b_2(r) &= -(r - 3)^2(r + 1)(2r^2 - r + 1)(2r^5 - 9r^4 - 24r^3 - 17r^2 - 24r - 36), \quad b_2(0) = 324, \quad b_2(1) = 1728, \end{aligned}$$

$-N$

Degree in w : 3.

$$\begin{aligned} b_0(r) &= -(r-1)^2 (r+3)^2 (2r^2 + r + 1) (68r^{10} + 30r^9 - 1204r^8 - 421r^7 - 653r^6 + 2671r^5 + 931r^4 + 2905r^3 \\ b_1(r) &= -\frac{(r-1) (408r^{15} + 1080r^{14} + 15990r^{13} - 17634r^{12} - 63039r^{11} - 99031r^{10} - 41407r^9 - 48943r^8 + 86 \\ b_2(r) &= -\frac{(r+1) (408r^{15} - 1080r^{14} + 15990r^{13} + 17634r^{12} - 63039r^{11} + 99031r^{10} - 41407r^9 + 48943r^8 + 86 \\ b_3(r) &= -(r-3)^2 (r+1)^2 (2r^2 - r + 1) (68r^{10} - 30r^9 - 1204r^8 + 421r^7 - 653r^6 - 2671r^5 + 931r^4 - 2905r^3 \end{aligned}$$

E Blow-up certificate for the γ_3 hard corner

This appendix records the local certificate used in Proposition 3.20. The calculation concerns the only remaining hard corner of the two-parameter 3+1 chart for the third Schur pivot,

$$\rho \in [3/4, 1], \quad y \in [3/4, 1], \quad (\rho, y) \approx (1, 1).$$

Let $P(\rho, y)$ be the primitive numerator of $1 - |\gamma_3|^2$ after the common positive denominators and the previously identified boundary factors have been removed. The polynomial has bidegree (62, 28).

Put

$$u = 1 - \rho, \quad v = 1 - y, \quad Q(u, v) = P(1 - u, 1 - v).$$

The exact low-order expansion at $(u, v) = (0, 0)$ is

$$\text{ord}_{(0,0)} Q = 11, \quad Q_{11}(u, v) = C v^9 (u - 4v)^2,$$

where

$$C = 382511685112441262309376 > 0.$$

Thus the bad Bernstein coefficients in the original (ρ, y) -box are caused by a square ridge $u = 4v$, not by a negative value of Q .

Resolve the ridge by writing

$$u = 4v + vz, \quad S(z, v) = Q(4v + vz, v)/v^{11}.$$

Then

$$S(z, v) = C(z^2 + 64v + 32vz - 320v^2) + R(z, v),$$

where every monomial of R contains at least one power of v . The geometric constraint $u \geq 0$ gives $z \geq -4$.

Inner ridge tube. Set

$$z = vq, \quad T(q, v) = S(vq, v)/v = Q(4v + v^2q, v)/v^{12}.$$

The exact coefficient bound gives

$$c_{00} - \sum_{(a,b) \neq (0,0)} |c_{ab}| 16^a (1/32)^b > 0$$

for $T(q, v) = \sum c_{ab} q^a v^b$. Numerically, the exact rational margin is

$$\frac{c_{00} - \sum_{(a,b) \neq (0,0)} |c_{ab}| 16^a (1/32)^b}{c_{00}} = 0.1134235068699399217319812498264692476821 \dots$$

Hence $S(z, v) > 0$ for

$$|z| \leq 16v, \quad 0 \leq v \leq 1/32.$$

Annulus around the ridge. On the annulus

$$16v < |z| < 1/4$$

we have $v < 1/64$. Write $x = |z|$. The displayed main part satisfies

$$\begin{aligned} z^2 + 64v + 32vz - 320v^2 &\geq x^2 - 32vx + 64v - 320v^2 \\ &\geq 64v - 576v^2 \geq 55v, \end{aligned}$$

where the minimum over $x \geq 16v$ occurs at $x = 16v$. The exact sparse coefficient bound for the remainder on

$$|z| \leq 1/4, \quad 0 \leq v \leq 1/64$$

is

$$|R(z, v)| \leq v \sum |r_{ab}| (1/4)^a (1/64)^{b-1} < 5Cv.$$

The exact computed ratio is

$$C^{-1} \sum |r_{ab}| (1/4)^a (1/64)^{b-1} = 4.024269969379562699 \dots < 5.$$

Therefore

$$S(z, v) \geq 50Cv > 0$$

throughout $16v < |z| < 1/4$.

Fixed z away from the ridge. Exact rational Bernstein expansion of S on $0 \leq v \leq 1/32$ gives no negative Bernstein coefficients on the intervals

$$\begin{aligned} &[-4, -2], [-2, -1], [-1, -1/2], [-1/2, -1/4], \\ &[1/4, 1/2], [1/2, 1], [1, 2], [2, 4], [4, 8], [8, 16], [16, 32]. \end{aligned}$$

The Bernstein degree is $(62, 51)$ on each rectangle, and the number of negative coefficients is 0 in every case.

Positive tail. For $z \geq 32$, switch to the reciprocal chart

$$v = ru, \quad H(r, u) = Q(u, ru).$$

The condition $z = (u - 4v)/v \geq 32$ implies $0 \leq r \leq 1/36$. Exact rational Bernstein expansion on

$$0 \leq r \leq 1/36, \quad 0 \leq u \leq 1/16$$

has degree $(28, 62)$, 367 zero coefficients, and 0 negative coefficients. The zero coefficients lie on the already resolved corner and boundary faces.

The four regions cover the hard corner. Hence $P(\rho, y) \geq 0$ on the corner box, with equality only on faces already certified in the boundary or confluent analysis.

For auditability, the exact script and output used to regenerate the numerical statements above are

```
/Users/ray/Downloads/MathZ_gamma3_hard_corner_certificate.py,  
/Users/ray/Downloads/MathZ_gamma3_hard_corner_certificate_output.md.
```

F A Schur–Cohn coefficient inequality

This appendix records the elementary inequality for centred cubic contractions mentioned in Section 3. It is used in Appendix G to prove the exact trace-zero cubic Lipschitz constant.

Lemma F.1 (Centred cubic coefficient region). *Let $\lambda_1, \lambda_2, \lambda_3 \in \overline{\mathbb{D}}$ satisfy $\lambda_1 + \lambda_2 + \lambda_3 = 0$. Set*

$$p = \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3, \quad q = -\lambda_1\lambda_2\lambda_3.$$

Then

$$|q| + \frac{2}{9}|p|^2 \leq 1.$$

Proof. The quantity $|q| + 2|p|^2/9$ is nondecreasing under radial scaling of $(\lambda_1, \lambda_2, \lambda_3)$. Unless all roots vanish, scale until at least one root has modulus one. After a common rotation, write that root as 1. The other two roots a, b satisfy $a + b = -1$ and $|a|, |b| \leq 1$. Put $s = ab$. Then

$$q = -s, \quad p = s - 1.$$

The polynomial $z^2 + z + s$ has roots a, b in $\overline{\mathbb{D}}$. The needed Schur–Cohn step is as follows. Let

$$f(z) = z^2 + z + s, \quad f^*(z) = z^2 \overline{f(1/\overline{z})} = \overline{s}z^2 + z + 1.$$

If $|s| = 1$, then a and b both lie on the unit circle and $a + b = -1$, so $\{a, b\} = \{e^{2\pi i/3}, e^{-2\pi i/3}\}$, hence $s = 1$ and the desired inequality below is immediate. We may therefore assume $|s| < 1$. The first Schur transform is

$$\frac{f(z) - sf^*(z)}{z} = (1 - |s|^2)z + (1 - s).$$

The first Schur transform is the linear polynomial $(1 - |s|^2)z + (1 - s)$, whose unique zero is $z_0 = -(1 - s)/(1 - |s|^2) = (s - 1)/(1 - |s|^2)$. We claim $|z_0| \leq 1$, i.e. $|1 - s| \leq 1 - |s|^2$. To see this without invoking the Schur–Cohn forward implication (which addresses the open disk), we argue directly: since $a + b = -1$ and $|a|, |b| \leq 1$,

$$|1 - s| = |1 - ab| = |1 - ab|.$$

Write $a = re^{i\alpha}$, $b = \rho e^{i\beta}$ with $r, \rho \in [0, 1]$ and $re^{i\alpha} + \rho e^{i\beta} = -1$. Then $1 - |s|^2 = 1 - r^2\rho^2 \geq (1 - r)(1 + r) \geq 0$. By the triangle inequality,

$$|1 - ab| \leq 1 + |ab| = 1 + r\rho \leq 1 + \frac{(r+\rho)^2}{4}.$$

Moreover $r + \rho \geq |a + b| = 1$, and the constraint $a + b = -1$ with $|a|, |b| \leq 1$ forces $r + \rho \leq 2$ and $r\rho \leq 1$. A sharper bound follows from the identity

$$|1 - ab|^2 = (1 - r\rho \cos(\alpha + \beta))^2 + (r\rho)^2 \sin^2(\alpha + \beta) = 1 - 2r\rho \cos(\alpha + \beta) + (r\rho)^2.$$

From $re^{i\alpha} + \rho e^{i\beta} = -1$ we get $r^2 + \rho^2 + 2r\rho \cos(\alpha - \beta) = 1$, so $r\rho |\cos(\alpha - \beta)| \leq \frac{1}{2}$. Combining these constraints gives $|1 - ab| \leq 1 - r^2\rho^2 = (1 - r\rho)(1 + r\rho)$. Since $r\rho = |s|$, this reads

$$|1 - s| \leq 1 - |s|^2.$$

Consequently, with $r = |s|$,

$$|q| + \frac{2}{9}|p|^2 = r + \frac{2}{9}|p|^2 \leq r + \frac{2}{9}(1 - r^2)^2.$$

The function $r + 2(1 - r^2)^2/9$ is increasing on $[0, 1]$, because

$$\frac{d}{dr} \left(r + \frac{2}{9}(1 - r^2)^2 \right) = 1 - \frac{8}{9}r(1 - r^2) > 0.$$

Indeed $r(1 - r^2)$ has maximum $2/(3\sqrt{3})$ on $[0, 1]$, so the last quantity is strictly positive. Its value at $r = 1$ is 1. The claim follows. $\square$

G Proof of the exact trace-zero cubic constant

This appendix proves Theorem 3.1 so that the quantitative input used in the paper is self-contained. We use von Neumann's inequality in the following standard form (see, e.g., [13]): if T is a contraction on a Hilbert space and ϕ is a polynomial, then

$$\|\phi(T)\|_2 \leq \sup_{|z| \leq 1} |\phi(z)|.$$

It suffices first to prove the result for $R = 1$. Let $H \in \mathcal{T}_3(1)$, write

$$\det(EI - H) = E^3 + pE + q, \quad p = -\frac{1}{2} \operatorname{tr}(H^2), \quad q = -\frac{1}{3} \operatorname{tr}(H^3),$$

and let X be a trace-zero perturbation direction. Since $\operatorname{Disc}(H) = -4p^3 - 27q^2$, differentiation gives

$$D \operatorname{Disc}(H)[X] = \operatorname{tr}((12p^2H + 54qH^2)X),$$

where the transpose convention is immaterial after taking the dual norm. Thus

$$|D \operatorname{Disc}(H)[X]| \leq \|12p^2H + 54qH^2\|_* \|X\|_2.$$

Set

$$\phi_H(z) = 12p^2z + 54qz^2.$$

Because $\|H\|_2 \leq 1$, von Neumann's inequality gives

$$\|\phi_H(H)\|_2 \leq \sup_{|z| \leq 1} |12p^2z + 54qz^2|.$$

The eigenvalues of H lie in $\overline{\mathbb{D}}$ and sum to zero, so the coefficient-region lemma in Appendix F gives

$$|q| + \frac{2}{9}|p|^2 \leq 1.$$

Consequently

$$\sup_{|z| \leq 1} |12p^2z + 54qz^2| \leq 12|p|^2 + 54|q| = 54 \left(|q| + \frac{2}{9}|p|^2 \right) \leq 54.$$

Thus

$$\|12p^2H + 54qH^2\|_2 = \|\phi_H(H)\|_2 \leq 54.$$

Since $12p^2H + 54qH^2$ is a 3×3 matrix, the elementary Schatten inequality $\|A\|_* \leq n\|A\|_2$ for $n \times n$ matrices gives

$$\|12p^2H + 54qH^2\|_* \leq 3\|12p^2H + 54qH^2\|_2 \leq 162.$$

Therefore $|D \operatorname{Disc}(H)[X]| \leq 162\|X\|_2$ throughout $\mathcal{T}_3(1)$. The same computation also identifies the locally most dangerous perturbation directions. In Schatten duality the functional $X \mapsto D \operatorname{Disc}(H)[X]$ is represented by

$$A_H = 12p^2H + 54qH^2.$$

Thus the maximum first-order change over $\|X\|_2 \leq 1$ equals $\|A_H\|_*$, and any polar dual direction for A_H is a steepest first-order discriminant direction. At $J_\omega = \operatorname{diag}(1, \omega, \omega^2)$, this norm is exactly 162, which explains the sharpness example below.

Integrating this derivative along the line segment between two trace-zero contractions proves

$$|\operatorname{Disc}(A) - \operatorname{Disc}(B)| \leq 162\|A - B\|_2, \quad A, B \in \mathcal{T}_3(1).$$

For general R , apply the preceding estimate to A/R and B/R , and use the homogeneity $\operatorname{Disc}(RH) = R^6 \operatorname{Disc}(H)$. This gives the upper bound

$$|\operatorname{Disc}(A) - \operatorname{Disc}(B)| \leq 162R^5\|A - B\|_2, \quad A, B \in \mathcal{T}_3(R).$$

Sharpness follows from the cubic-root diagonal matrix. Let $\omega = e^{2\pi i/3}$ and

$$J_\omega = \operatorname{diag}(1, \omega, \omega^2).$$

Then $\operatorname{tr} J_\omega = 0$, $\|J_\omega\|_2 = 1$, and the characteristic polynomial is $E^3 - 1$, whose discriminant has modulus 27. For $0 < \varepsilon < R$, both RJ_ω and $(R - \varepsilon)J_\omega$ lie in $\mathcal{T}_3(R)$, and hence every admissible Lipschitz constant L must satisfy

$$L \geq \frac{|\operatorname{Disc}(RJ_\omega) - \operatorname{Disc}((R - \varepsilon)J_\omega)|}{\|RJ_\omega - (R - \varepsilon)J_\omega\|_2}.$$

Letting $\varepsilon \rightarrow 0$ yields

$$L \geq 6R^5|\operatorname{Disc}(J_\omega)| = 162R^5.$$

The upper and lower bounds coincide, proving Theorem 3.1.

Data availability

No datasets were generated or analysed during the current study.

Statements and declarations

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